Basic Models for Classification

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Outline

- Two Simple Approaches
- The Curse of Dimensionality
- Linear Discriminant Analysis
- Naive Bayes Classifier
- Logistic Regression

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Formulation

- Observe a collection of i.i.d. **training data**

$$(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n) \text{ from } D.$$

- **Predictors** (covariates, inputs, features):

$$x_i = (x_{i1}, \ldots, x_{ip}).$$

- **Response** (class, label):

$$y_i \in C, C = \{c_1, \ldots, c_K\}.$$

- When there are only two classes (a binary classification), is often coded as 0/1 or 1/-1.

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The Regression Approach

- For a binary classification, let y be 0/1 coded. Fit a regression $\hat{f}(x)$, and let the classification rule be

$$\hat{y} = \begin{cases} 1 & \text{if } \hat{f}(x) > 0.5, \\ 0 & \text{if } \hat{f}(x) \leq 0.5. \end{cases}$$

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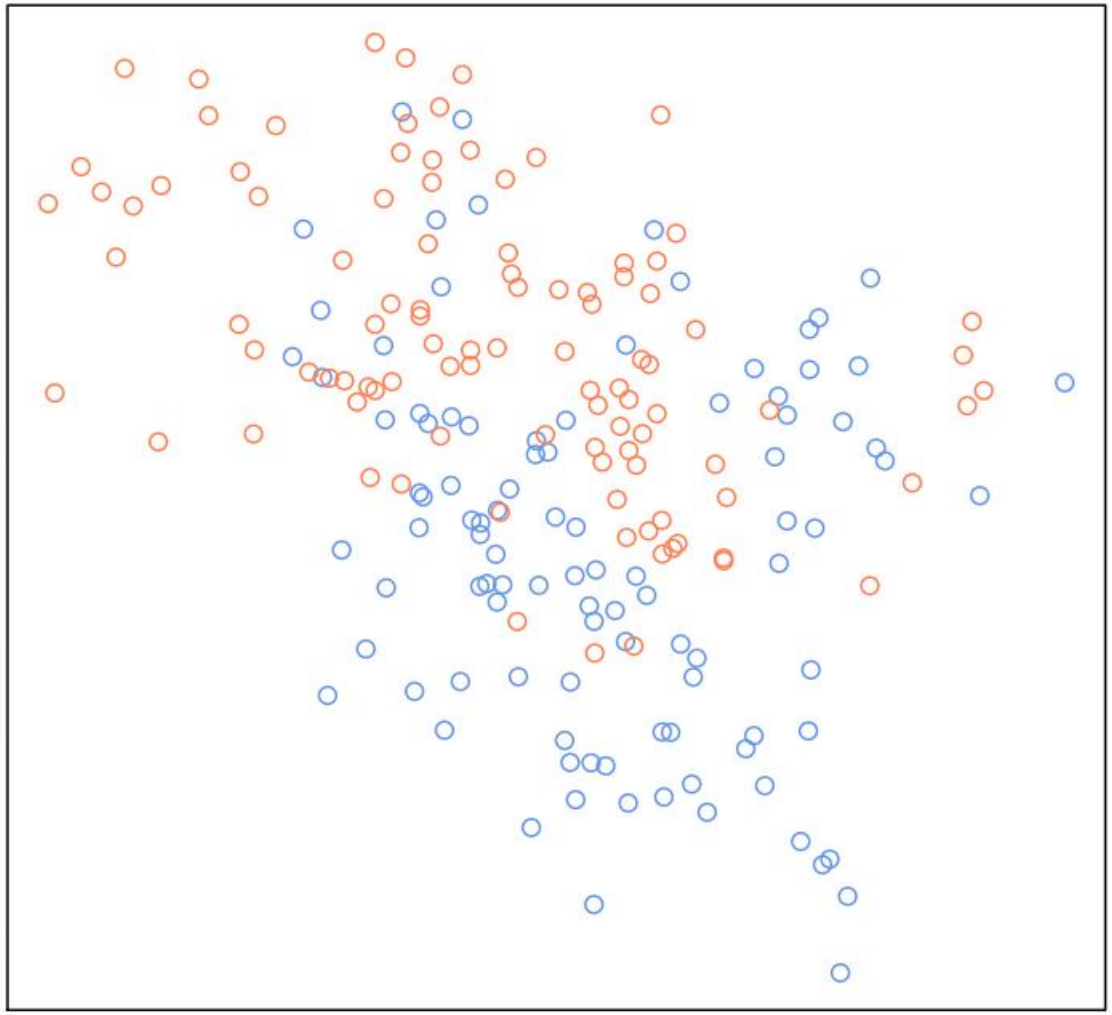
The Regression Approach

- More generally, with K classes, let $y_k = I_{\{y=k\}}$. Fit regression $y_k = \hat{f}_k(x)$, for $k = 1, \ldots, K$, and

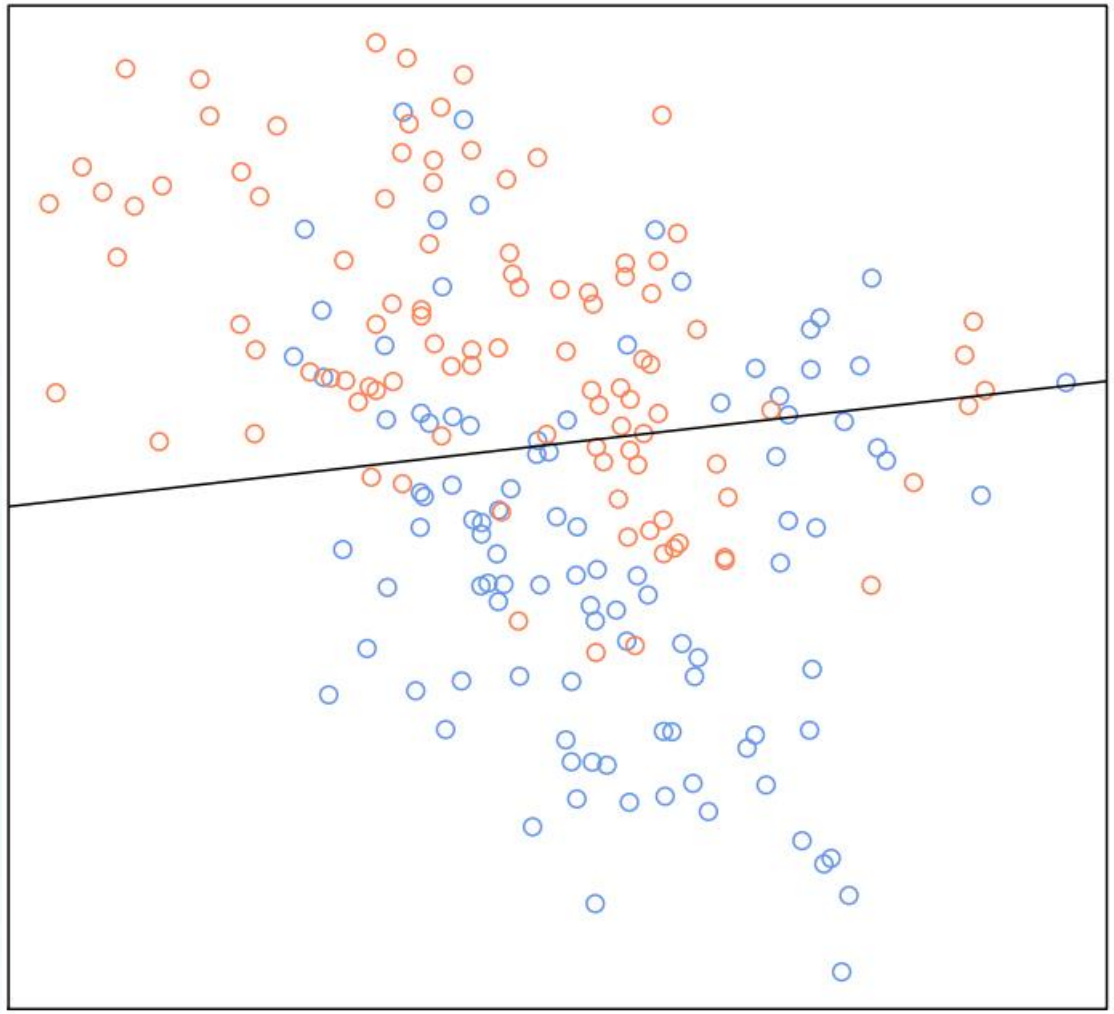
$$\hat{y} = \underset{k}{\operatorname{argmax}} \hat{f}_k(x).$$

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The Regression Approach



The Regression Approach



Metrics for Accuracy

- Accuracy:

$$P(\hat{Y} = Y) \text{ or } \frac{\sum_{i=1}^{n} I(y_i = \hat{y}_i)}{n}$$

- Error rate (misclassification rate):

$$P(\hat{Y} \neq Y) \text{ or } \frac{\sum_{i=1}^{n} I(y_i \neq \hat{y}_i)}{n}$$

- For binary classifications:
  - Prevalence:

$$P(Y = 1) \text{ or } \frac{\sum_{i=1}^{n} y_i}{n}$$

Metrics for Accuracy

- True positive rate (TPR, sensitivity, recall):

$$P\left(\hat{Y}=1|Y=1\right) \text{ or } \frac{\sum_{i=1}^{n} I(\hat{y}_i=1)\cdot I(y_i=1)}{\sum_{i=1}^{n} I(y_i=1)}$$

- False positive rate (FPR):

$$P\left(\hat{Y}=1|Y=0\right) \text{ or } \frac{\sum_{i=1}^{n} I(\hat{y}_i=1)\cdot I(y_i=0)}{\sum_{i=1}^{n} I(y_i=0)}$$

- True negative rate (TNR, specificity, SPC):

$$P\left(\hat{Y}=0|Y=0\right) \text{ or } \frac{\sum_{i=1}^{n} I(\hat{y}_i=0)\cdot I(y_i=0)}{\sum_{i=1}^{n} I(y_i=0)}$$

- False negative rate (FNR):

$$P\left(\hat{Y}=0|Y=1\right) \text{ or } \frac{\sum_{i=1}^{n} I(\hat{y}_i=0)\cdot I(y_i=1)}{\sum_{i=1}^{n} I(y_i=1)}$$

Metrics for Accuracy

- Positive predictive value (PPV, precision):

$$P\left(Y=1|\hat{Y}=1\right) \text{ or } \frac{\sum_{i=1}^{n} I(\hat{y}_i=1)\cdot I(y_i=1)}{\sum_{i=1}^{n} I(\hat{y}_i=1)}$$

- False discovery rate (FDR):

$$P\left(Y=0|\hat{Y}=1\right) \text{ or } \frac{\sum_{i=1}^{n} I(\hat{y}_i=1)\cdot I(y_i=0)}{\sum_{i=1}^{n} I(\hat{y}_i=1)}$$

- Negative predictive value (NPV):

$$P\left(Y=0|\hat{Y}=0\right) \text{ or } \frac{\sum_{i=1}^{n} I(\hat{y}_i=0)\cdot I(y_i=0)}{\sum_{i=1}^{n} I(\hat{y}_i=0)}$$

- False omission rate (FOR):

$$P\left(Y=1|\hat{Y}=0\right) \text{ or } \frac{\sum_{i=1}^{n} I(\hat{y}_i=0)\cdot I(y_i=1)}{\sum_{i=1}^{n} I(\hat{y}_i=0)}$$

Metrics for Accuracy

- F1 score (or F-score):

$$\text{F1 score} = 2 \cdot \frac{\text{precision} \cdot \text{recall}}{\text{precision} + \text{recall}}$$

两者的调和平均值

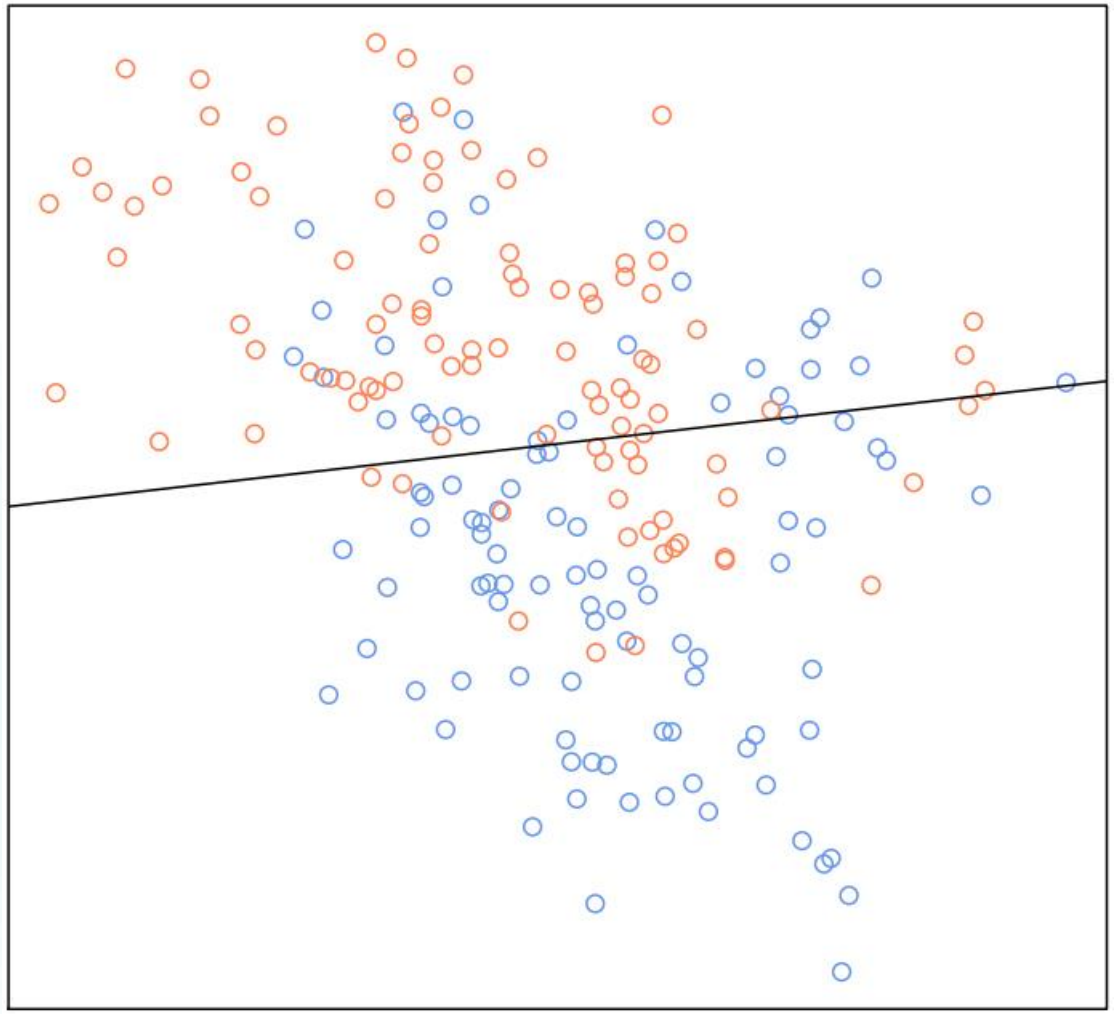
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Extension to multi-class

- Micro-averaging: samples have equal weights 指向样本，每个样本的权重相同
  - Recall = TP_sum / TrueLabel_sum = TP_sum / N = accuracy
  - Precision = TP_sum / PredictLabel_sum = TP_sum / N = accuracy
  - F score = Harmonic mean of the micro-average recall and precision = accuracy
- Macro-averaging: classes have equal weights 指向类，每个类的权重相同
  - Recall = mean of per class recall
  - Precision = mean of per class precision
  - F score
    - Simple average of per class F score
    - Harmonic mean of the macro-average recall and precision

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The Regression Approach



Metrics for Accuracy

Prevalence	Accuracy	Error rate	F1 score
0.5	0.73	0.27	0.7378641

True positive rate	False positive rate	True negative rate	False negative rate
0.76	0.3	0.7	0.24

Positive predictive value	False discovery rate	Negative predictive value	False omission rate
0.7169811	0.2830189	0.7446809	0.2553191

(Training Set)

Metrics for Accuracy

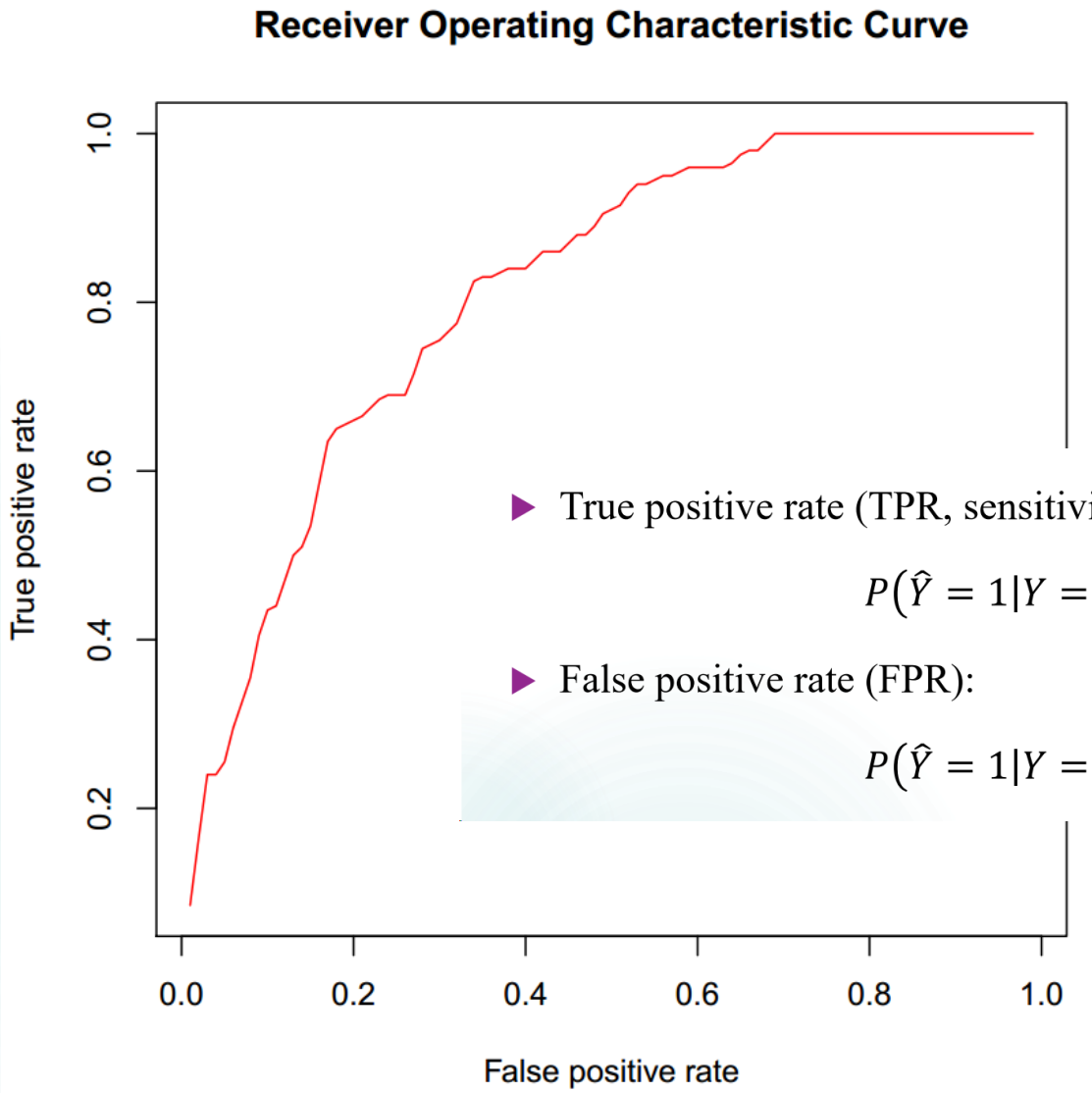
- The threshold for cutoff does not have to be 0.5!

$$\hat{y} = \begin{cases} 1, \text{if } \hat{f}(x) > s, \\ 0, \text{if } \hat{f}(x) \leq s. \end{cases}$$

- And the above metrics for accuracy will be dependent on s.
- Is there a metric that is independent of the cutoff threshold?

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ROC Curve



- True positive rate (TPR, sensitivity, recall):

$$P(\hat{Y}=1|Y=1) \text{ or } \frac{\sum_{i=1}^{n} I(\hat{y}_i=1)\cdot I(y_i=1)}{\sum_{i=1}^{n} I(y_i=1)}$$

- False positive rate (FPR):

$$P(\hat{Y}=1|Y=0) \text{ or } \frac{\sum_{i=1}^{n} I(\hat{y}_i=1)\cdot I(y_i=0)}{\sum_{i=1}^{n} I(y_i=0)}$$

AUC

更具体地：

$$\mathrm{AUC} = \int_0^1 \mathrm{TPR}(\mathrm{FPR})\, d\mathrm{FPR}$$

- 这相当于**在所有可能的FPR值下计算模型的平均TPR。**
- AUC越高，意味着在各种FPR情况下，TPR都相对较高，即模型能有效区分正负样本。

2. "The expected true positive rate if the ranking is split just before a uniformly drawn random negative."

（AUC等于：如果选择某个负样本作为阈值，那么模型在此阈值下的期望真阳性率（TPR）。）

- 设想在模型的预测分数排序中，**随机选取一个负样本，并把它当作分类阈值。**
- 在这个阈值下，计算所有正样本的**真阳性率（TPR）**：

$$TPR = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}}$$

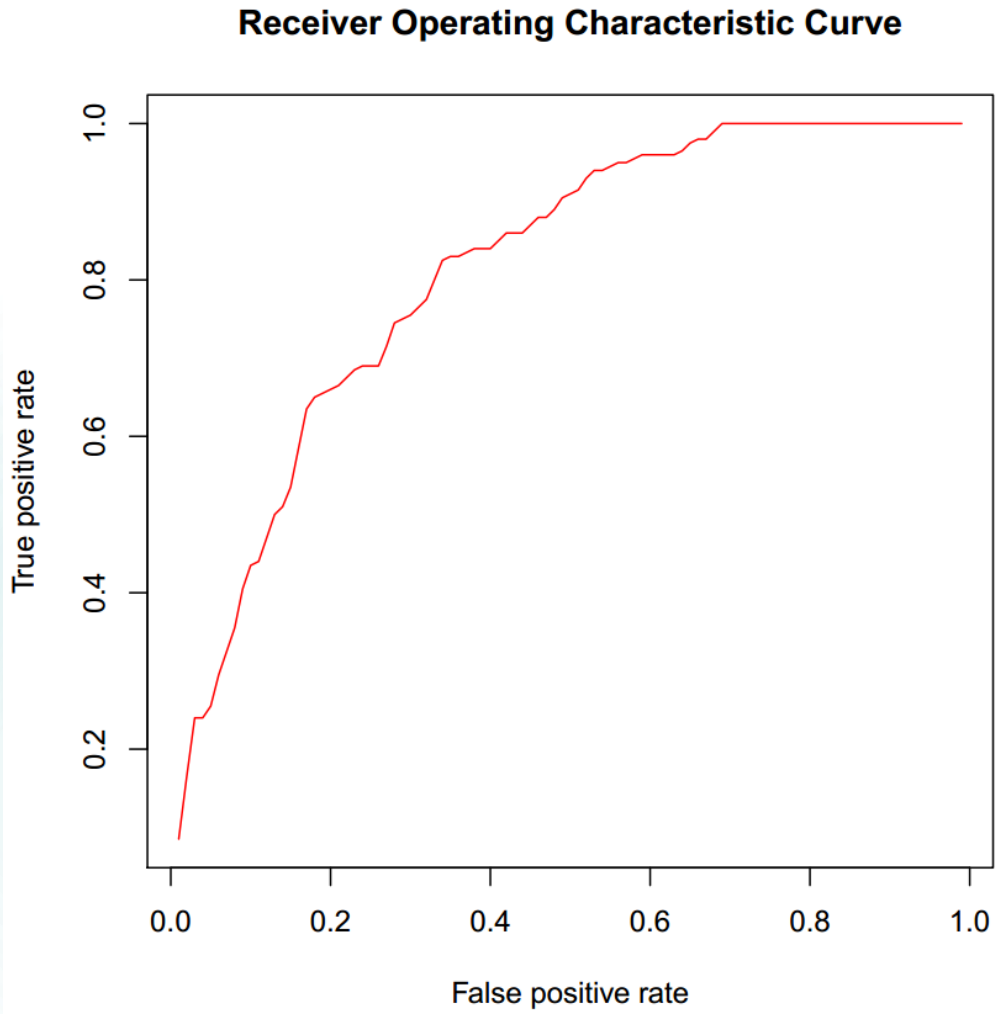
- 由于负样本是随机选的，这个TPR在所有可能阈值下的**期望值**，正好就是AUC。

Area under the ROC curve

- The probability that a uniformly drawn random positive is ranked before a uniformly drawn random negative.
- The expected true positive rate if the ranking is split just before a uniformly drawn random negative.
- The expected proportion of positives ranked before a uniformly drawn random negative.
- The expected proportion of negatives ranked after a uniformly drawn random positive.
- 1 minus the expected false positive rate if the ranking is split just after a uniformly drawn random positive.

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Metrics for Accuracy



AUC
0.812

ROC Curve

Where did the name come from?

It came from a field called "Signal Detection Theory" developed during World War II for the analysis of radar images. Radar operators had to decide whether a blip on the screen represented an enemy target, a friendly ship, or just noise. Signal detection theory measures the ability of radar receiver operators to make these important distinctions. Their ability to do so was called the Receiver Operating Characteristics.

The Nearest-Neighbors Approach

- The k-nearest neighbor fit is defined as

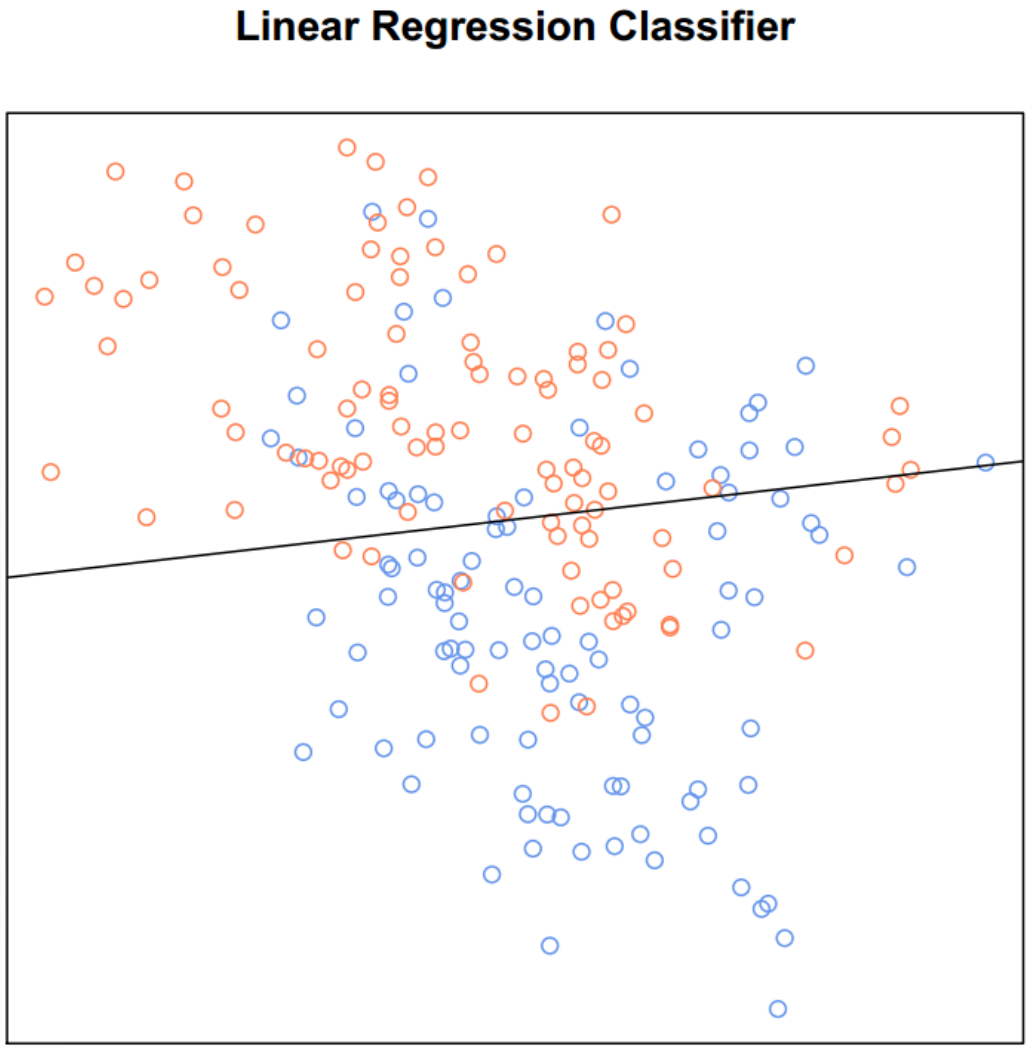
$$\hat{f}(x) = \frac{1}{k} \sum_{x_i \in N_k(x)} y_i \,,$$

where $N_k(x)$ is the neighborhood of x defined by the closest points x_i in the training sample.

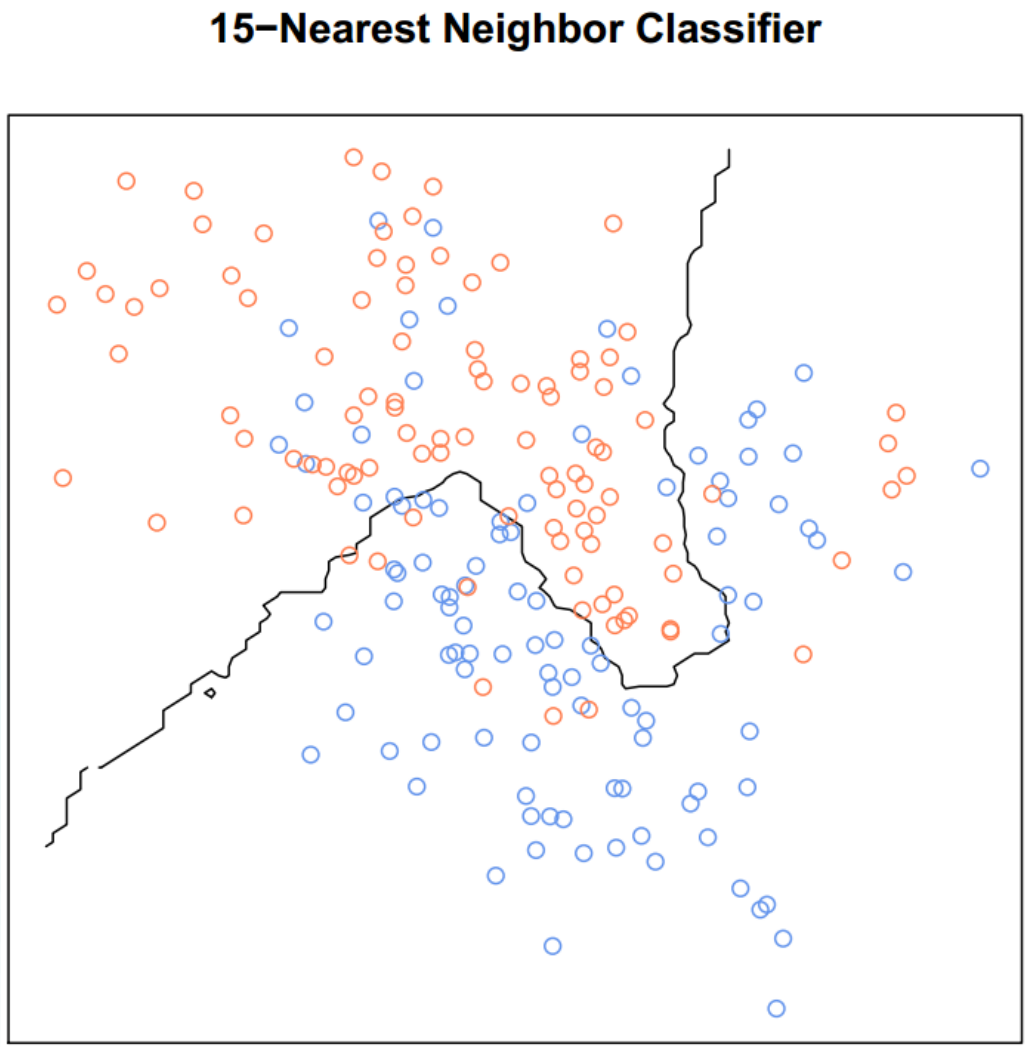
$$\hat{y} = \begin{cases} 1, \text{if } \hat{f}(x) > s, \\ 0, \text{if } \hat{f}(x) \leq s. \end{cases}$$

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The Nearest-Neighbors Approach



The Nearest-Neighbors Approach



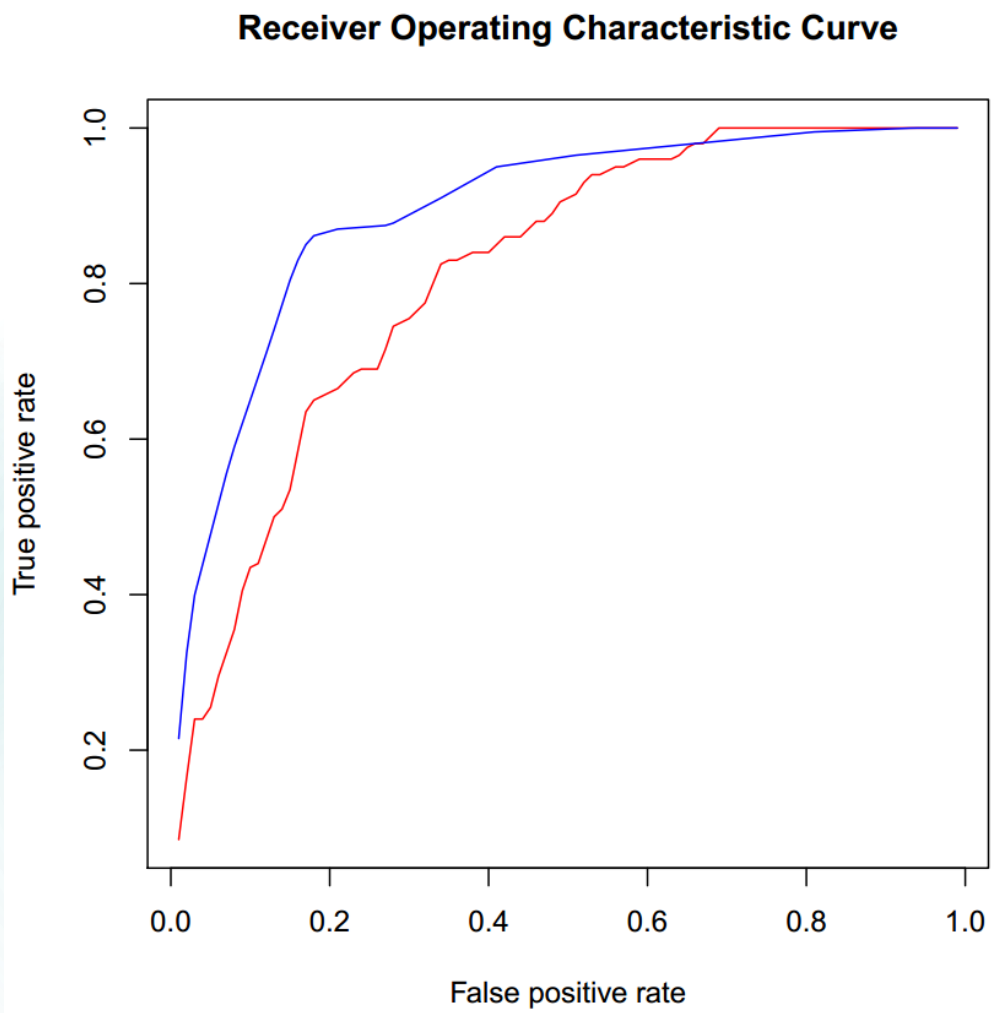
Accuracy

Prevalence	Accuracy	Error rate	F1 score
0.5	0.845	0.155	0.8487805

True positive rate	False positive rate	True negative rate	False negative rate
0.87	0.18	0.82	0.13

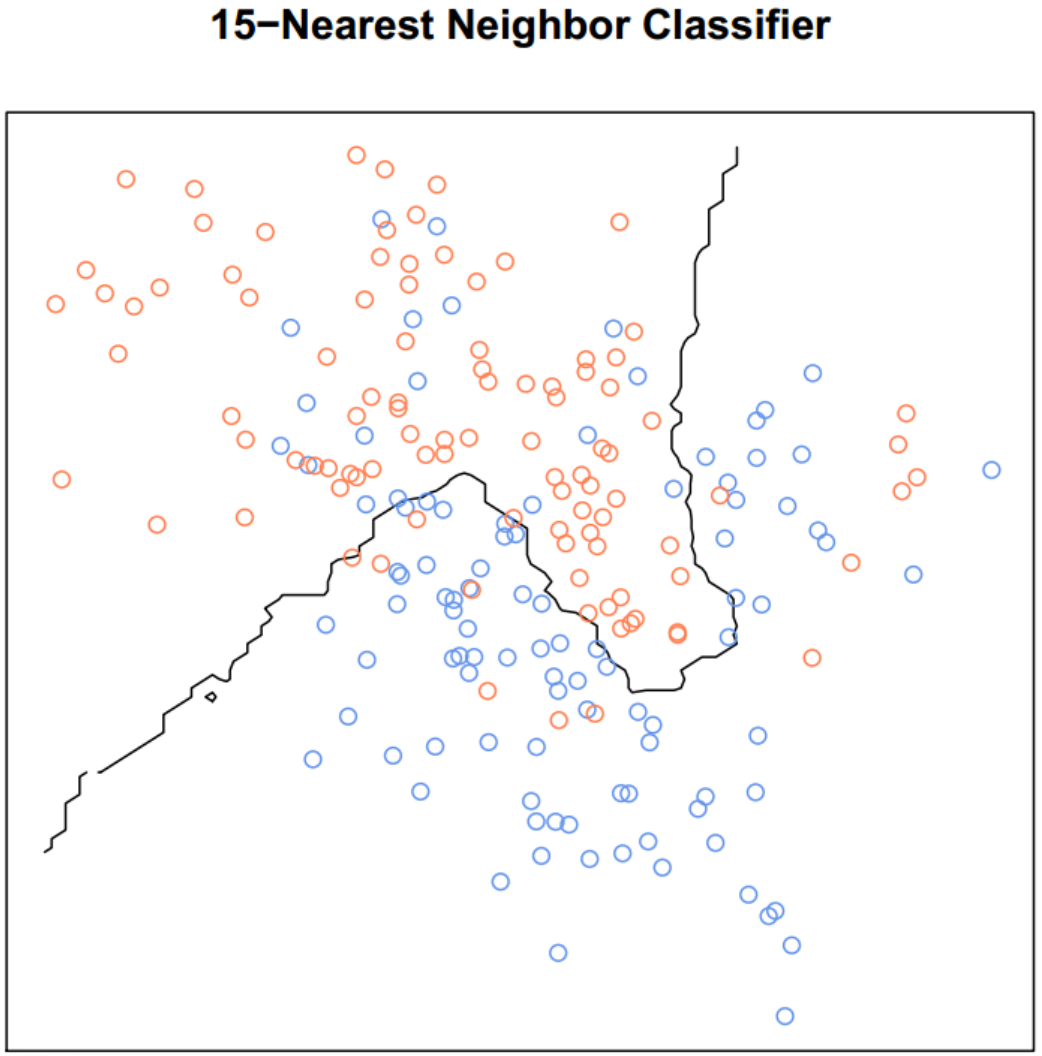
Positive predictive value	False discovery rate	Negative predictive value	False omission rate
0.8285714	0.1714286	0.8631579	0.1368421

Accuracy

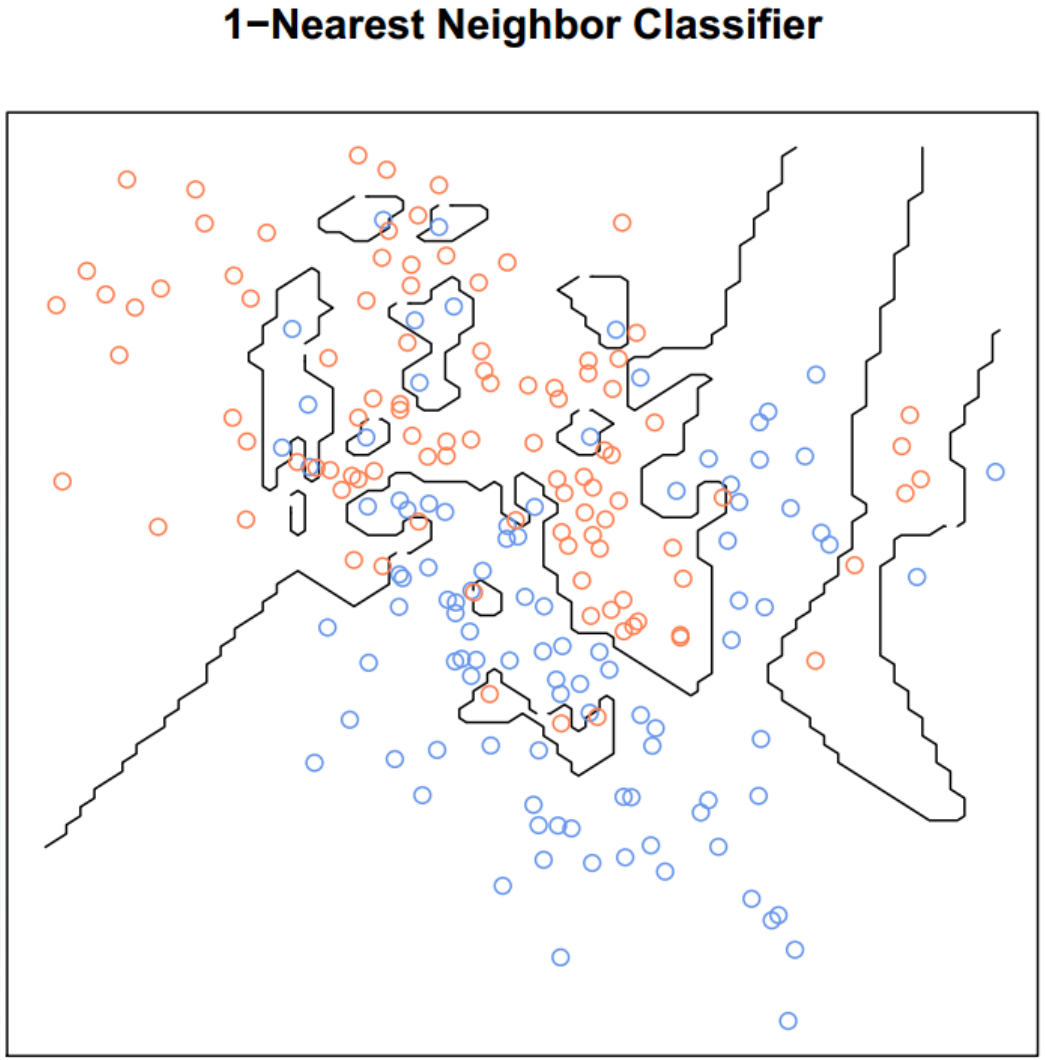


OLS AUC
0.812
KNN AUC
0.8934

The Nearest-Neighbors Approach



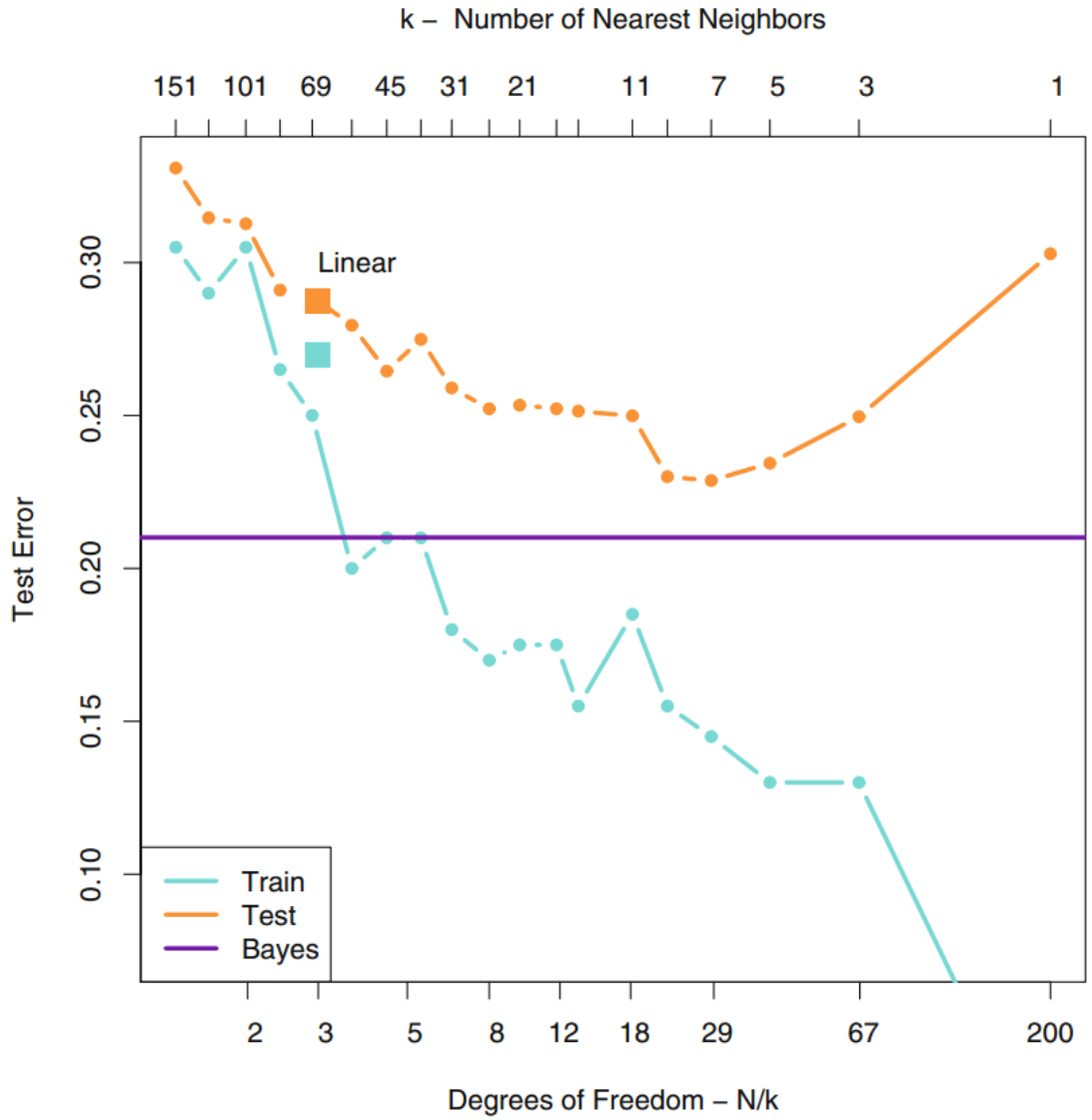
The Nearest-Neighbors Approach



Overfitting

- *Overfitting* describes the state where the model corresponds too closely to the training samples but fails to extract the underlying pattern for generalization.
- Overfitting can occur when a model is excessively complex, such as having too many parameters relative to the number of observations.
- In the past, large models were almost synonymous to overfitting, which is not true. However, fitting large models with small data certainly does not bode well.
- Whether overfitting happens is affected by sample size, sample/feature representativeness, and training techniques.

Cross-Validation



Bayes:

Statistical Decision Theory

- The expected loss

$$E(L) = E_X\left(L\left(Y, \hat{y}(X)\right)\right)$$
$$= E_X \sum_{k=1}^{K} L\left(k, \hat{y}(X)\right) \cdot P(Y = k|X).$$

- It suffices to optimize pointwise:

$$\hat{y}(x) = \operatorname{argmin}_j \sum_{k=1}^{K} L(k, j) \cdot P(Y = k|X = x).$$

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Statistical Decision Theory

- With 0-1 loss, the decision simplifies to:

$$\hat{y}(x) = \operatorname{argmin}_j \sum_{k=1}^{K} L(k,j) \cdot P(Y = k | X = x)$$

$$= \operatorname{argmin}_j P(Y \neq j | X = x)$$

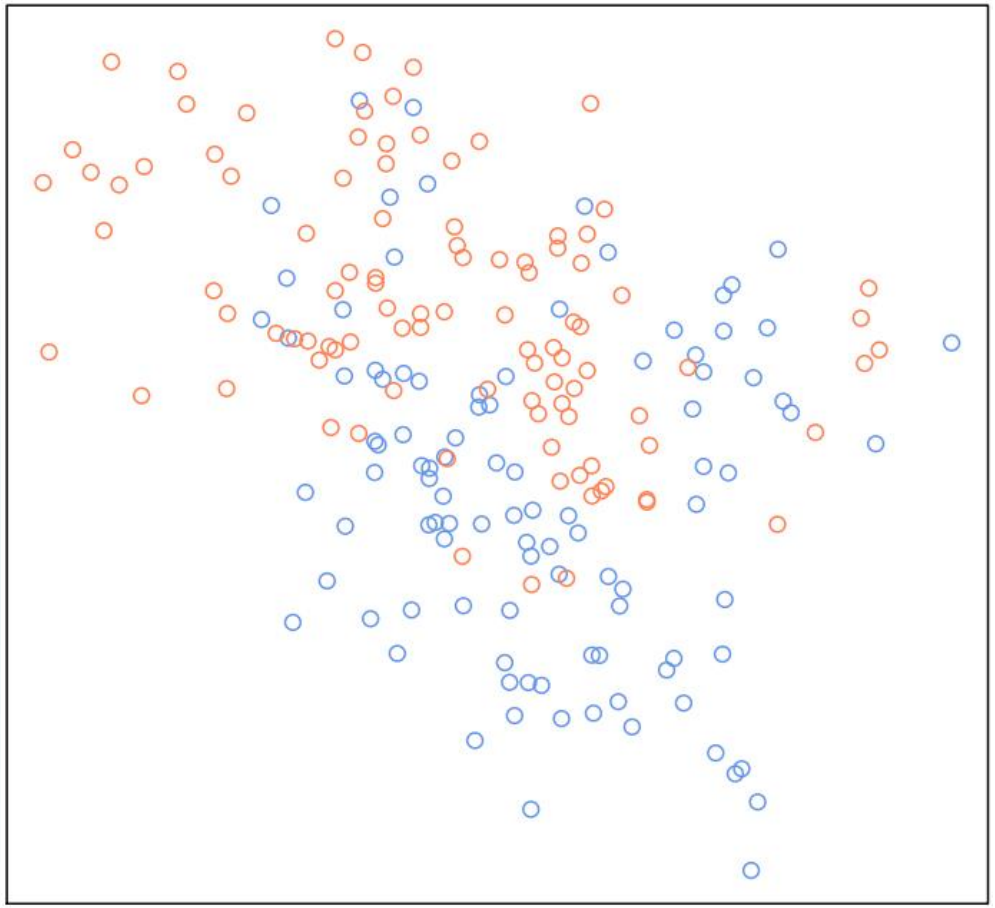
$$= \operatorname{argmax}_j P(Y = j | X = x).$$

- The above is known as the *Bayes (optimal) classifier*.
- The error rate of the Bayes classifier is the *optimal rate*.

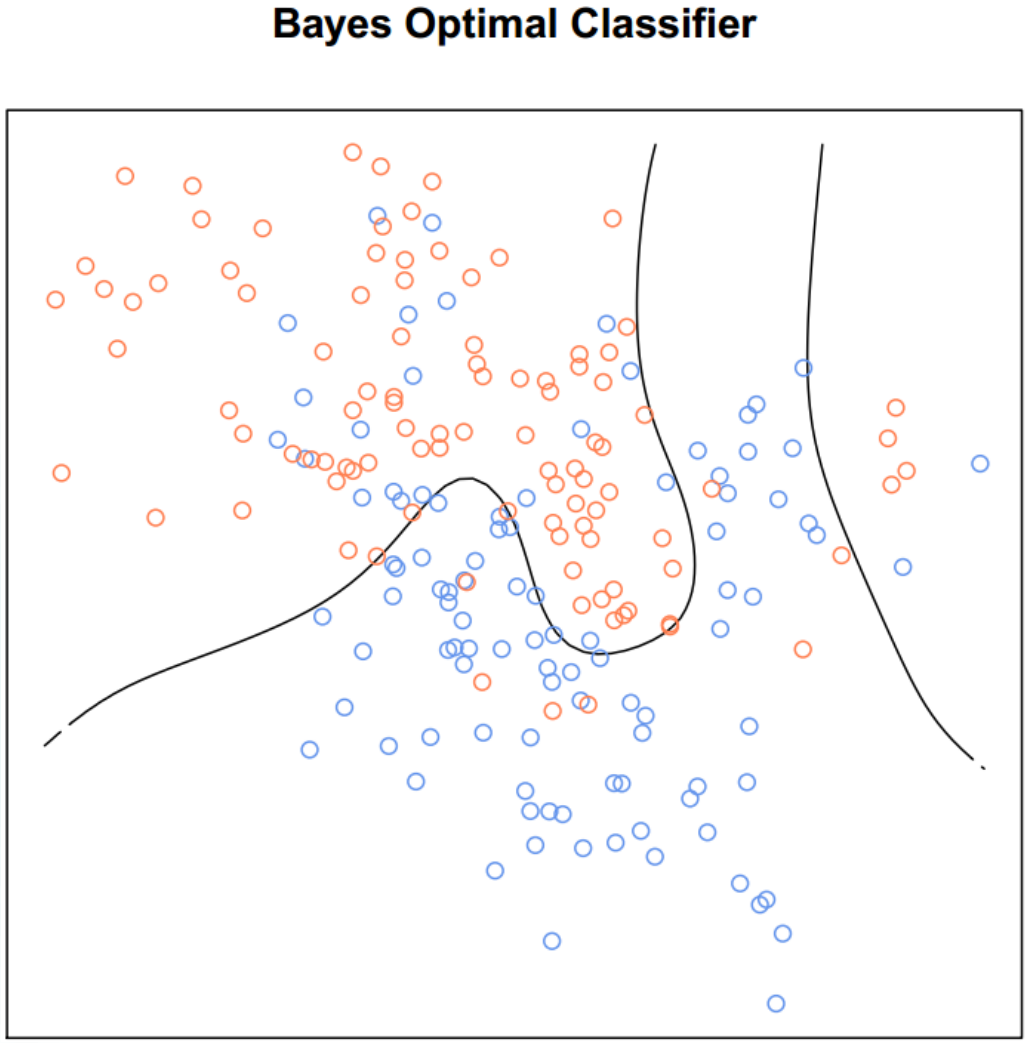
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Statistical Decision Theory

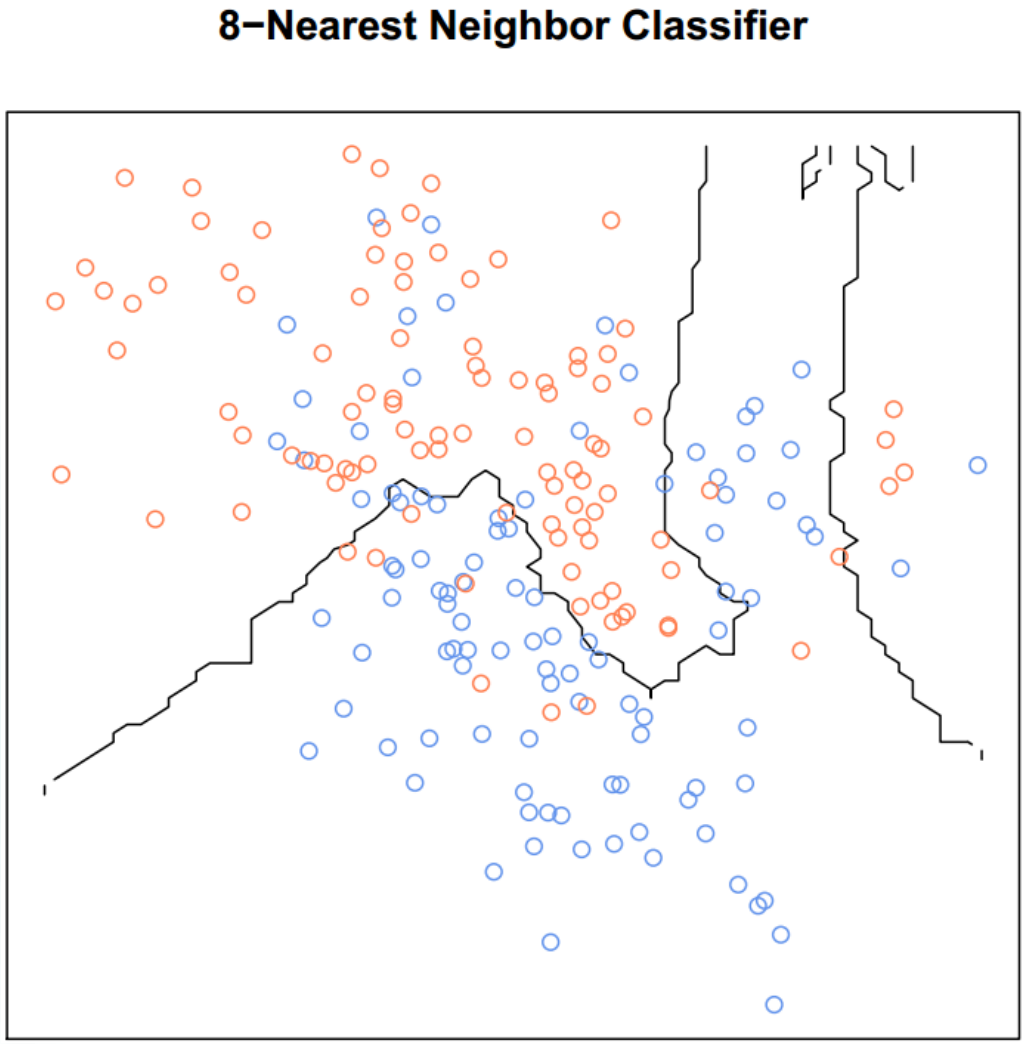
- The underlying distribution: A mixture of multiple normal distributions.



Statistical Decision Theory



Statistical Decision Theory



Is KNN Ideal?

- It would seem that KNN is an ideal method, especially when we have large samples.
- However, this intuition breaks down in higher dimensions.

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The Curse of Dimensionality

The Curse of Dimensionality

- The term “curse of dimensionality” was originally coined by Bellman to describe the difficulty in solving high dimensional dynamic programs.

- The term was later used by statisticians to describe a different issue in machine learning.

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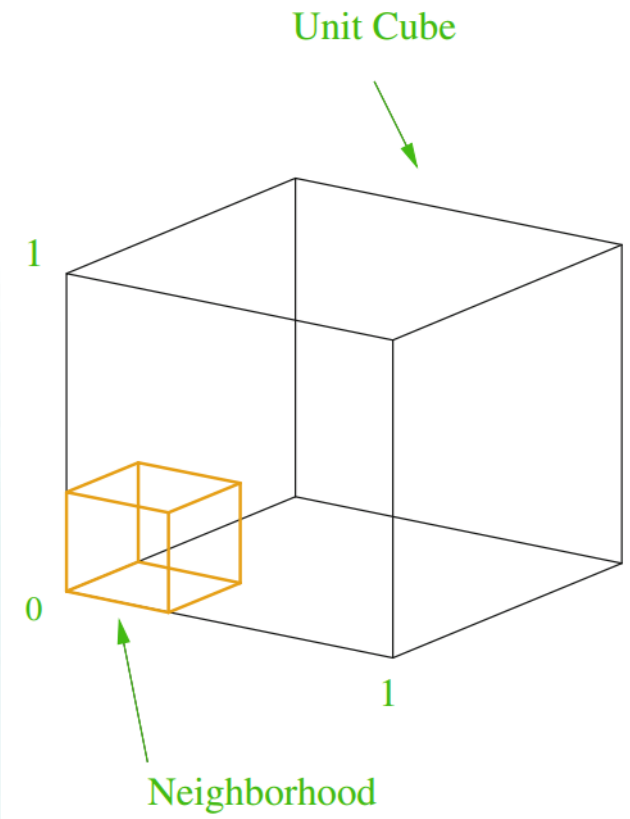
The Curse of Dimensionality

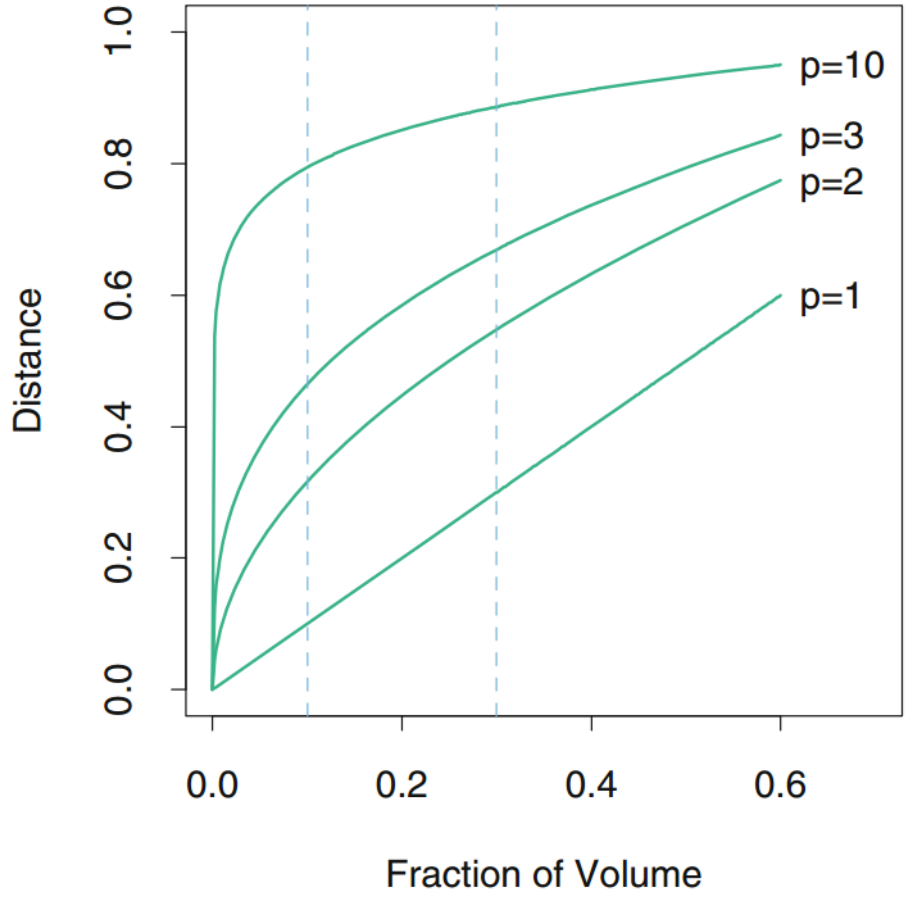
- Consider a p-dimensional unit hypercube.
  - Each feature ranges from 0 to 1.
  - The unit hypercube represent the complete feature space.
  - Samples are uniformly scattered in the feature space.
  - To cover r proportion of the samples as neighbors, one needs a subcube with edge length

$$d = r^{1/p}.$$

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The Curse of Dimensionality





- For $r = 0.01\%$ and $p = 100$, $d = 0.912$.

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The Curse of Dimensionality

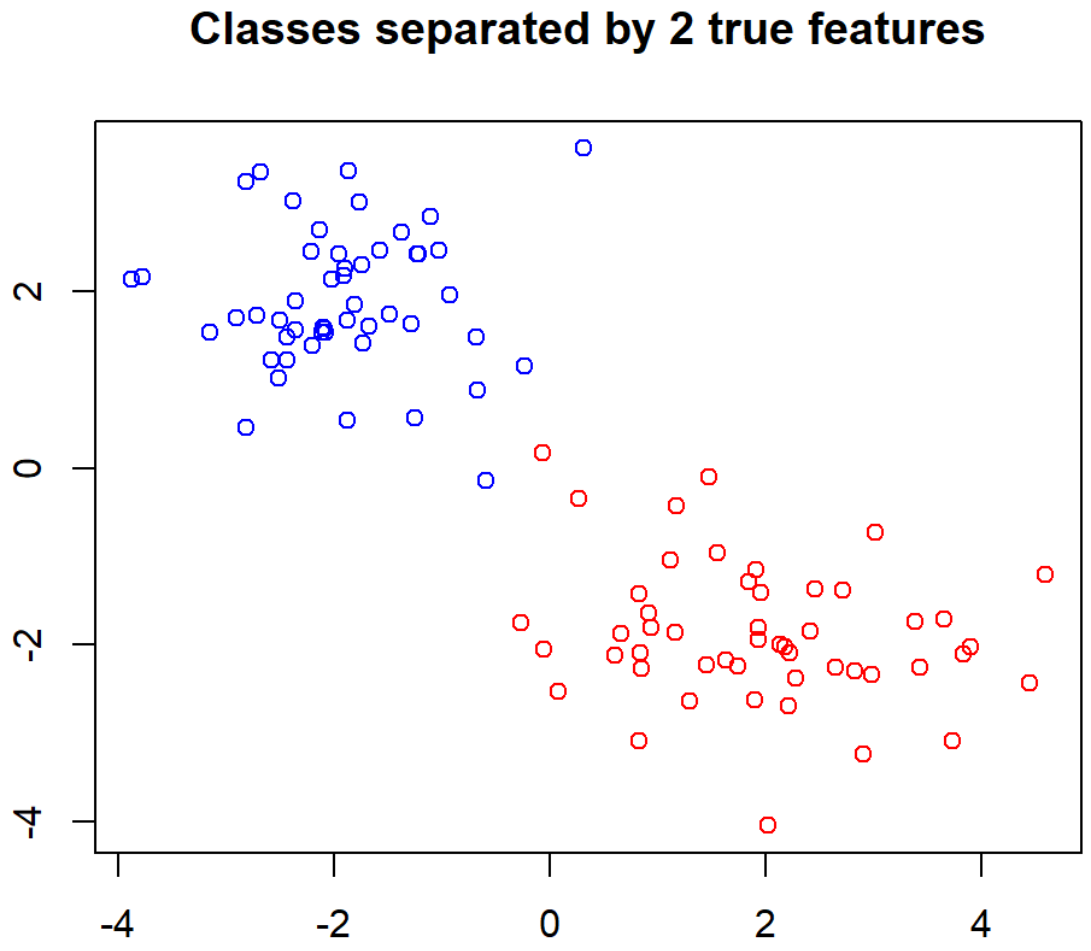
- Consider N data points uniformly distributed in a p-dimensional unit ball centered at the origin.
- The median distance from the origin to the closest data point is given by the expression

$$d(p,N) = \left(1 - \left(\frac{1}{2}\right)^{1/N}\right)^{1/p} .$$

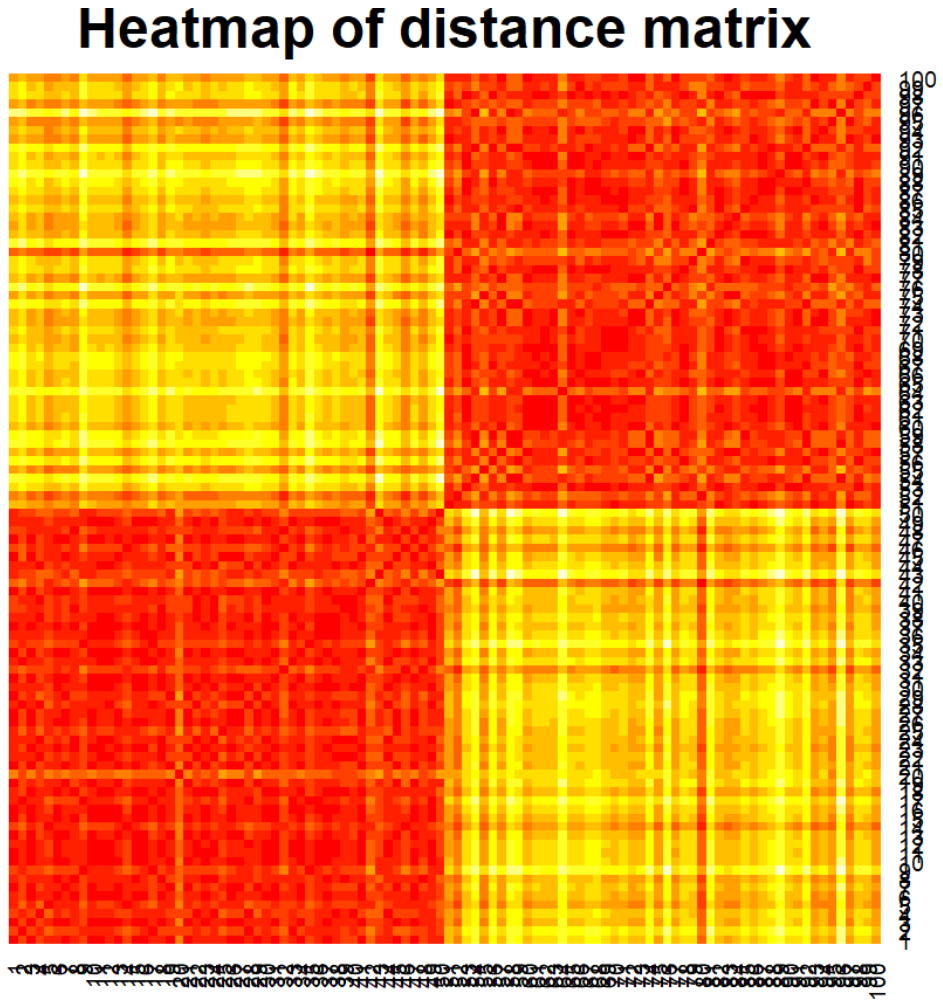
- For $N = 10000$ and $p = 100$, $d(p,N) = 0.909$.

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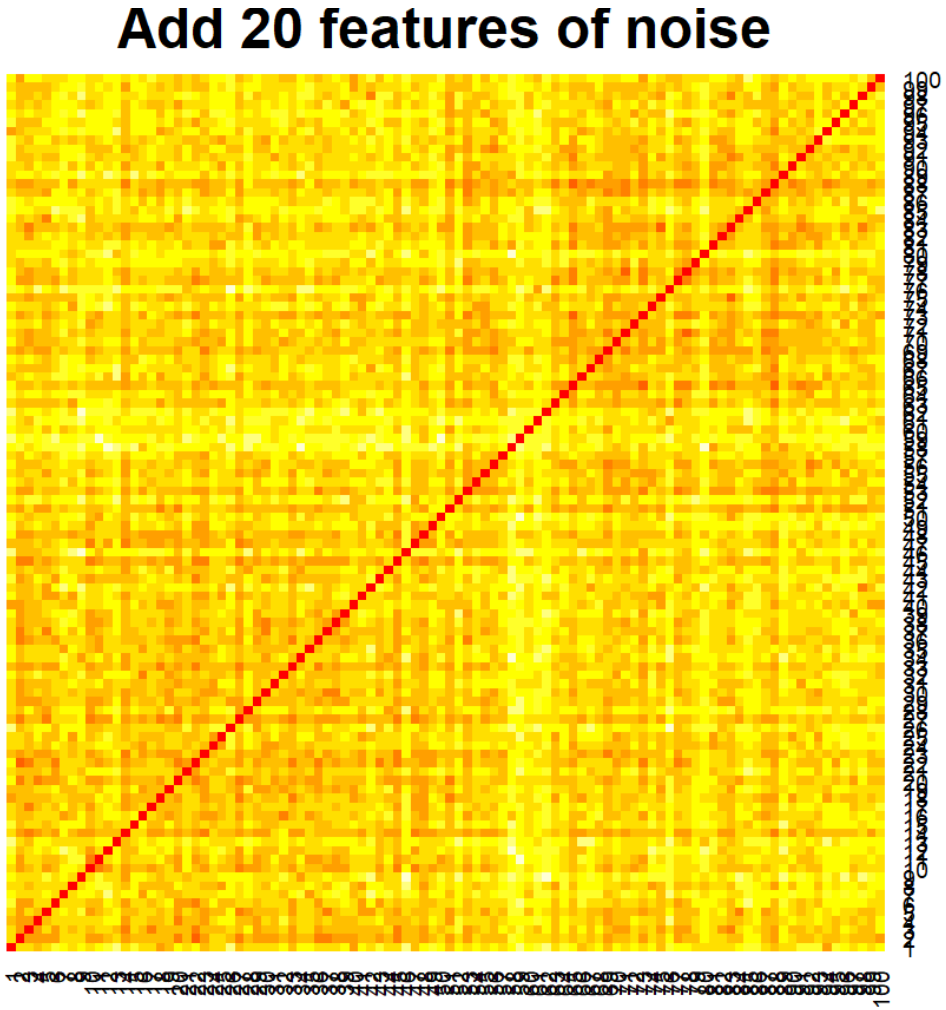
The Curse of Dimensionality



The Curse of Dimensionality



The Curse of Dimensionality



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Models Based on Densities

Models Based on Densities

- Decision theory for classification tells us that we need to know the class posteriors $P(Y|X)$ for optimal classification.

- Suppose $f_k(x)$ is the class-conditional density of X in class k.

- Let π_k be the prior probability of class k, with $\sum_{k=1}^{K} \pi_k = 1$.

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Models Based on Densities

- By Bayes theorem,

$$P(Y = k|X = x) = \frac{f_k(x)\pi_k}{\sum_{l=1}^{K} f_l(x)\pi_l}.$$

Thus,

$$\hat{y}(x) = \underset{k}{\operatorname{argmax}} f_k(x)\pi_k.$$

- Linear/quadratic discriminant analysis
- Naive Bayes

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Linear Discriminant Analysis

- Suppose that we model each class density as multivariate Gaussian

$$f_k(x) = \frac{1}{(2\pi)^{p/2}|\Sigma_k|^{1/2}} \exp\left\{-\frac{1}{2}(x-\mu_k)^T \Sigma_k^{-1}(x-\mu_k)\right\}.$$

In comparing two classes k and l, it is sufficient to look at the log-ratio

$$\log\frac{P(k|x)}{P(l|x)} = \log\frac{f_k(x)\pi_k}{f_l(x)\pi_l}.$$

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Linear Discriminant Analysis

- Linear discriminant analysis (LDA) arises in the special case when we assume that the classes have a common covariance matrix $\Sigma_k = \Sigma$, for any k. Thus

$$\log\frac{P(k|x)}{P(l|x)} = \log\frac{f_k(x)\pi_k}{f_l(x)\pi_l}$$

$$= \log\frac{\pi_k}{\pi_l} - \frac{1}{2}(\mu_k + \mu_l)^T\Sigma^{-1}(\mu_k - \mu_l) + x^T\Sigma^{-1}(\mu_k - \mu_l).$$

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Linear Discriminant Analysis

- Equivalently, we can use the *linear discriminant functions*

$$\delta_k(x) = \log \pi_k + x^T \Sigma^{-1} \mu_k - \frac{1}{2} \mu_k^T \Sigma^{-1} \mu_k,$$

and

$$\hat{y}(x) = \underset{k}{\operatorname{argmax}} \delta_k(x).$$

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Linear Discriminant Analysis

- In practice we do not know the parameters of the Gaussian distributions, and will need to estimate them using our training data:

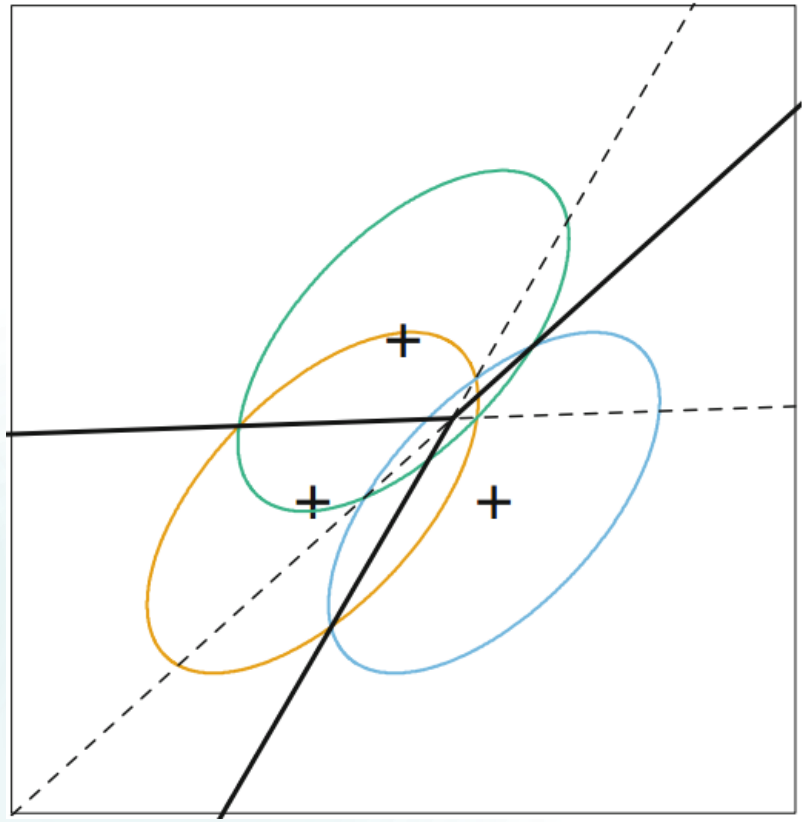
- $\hat{\pi}_k = N_k/N$, where N_k is the number of class-k observations;

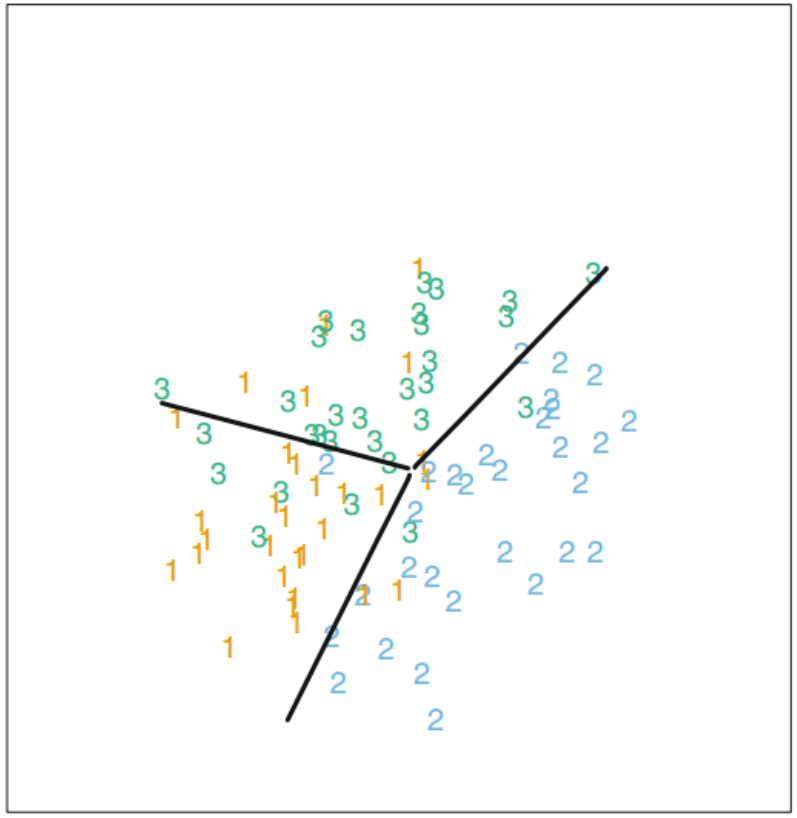
- $\hat{\mu}_k = \sum_{y_i=k} x_i/N_k$;

- $\hat{\Sigma} = \frac{\sum_{k=1}^{K}\sum_{y_i=k}(x_i-\hat{\mu}_k)(x_i-\hat{\mu}_k)^T}{N-K}$.

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Linear Discriminant Analysis





- Left: Underlying distributions and Bayes classification boundaries.
- Right: LDA classification with 30 samples from each class.

Linear Discriminant Analysis

- With $K = 2$, the coefficient vector from LDA is proportional to that from the least squares method.
- The intercepts are the same when $N_1 = N_2$.
- Thus the Gaussian assumption is not strictly required.
- With more than two classes, LDA is not the same as the linear regression approach.

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Quadratic Discriminant Analysis

- Two ways to obtain nonlinear classification boundaries from LDA:
  - Add polynomial features, such as X_i^2 and $X_i X_j$;
  - Relax the common covariance matrix $\Sigma_k = \Sigma$ constraint, and LDA becomes QDA.

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Quadratic Discriminant Analysis

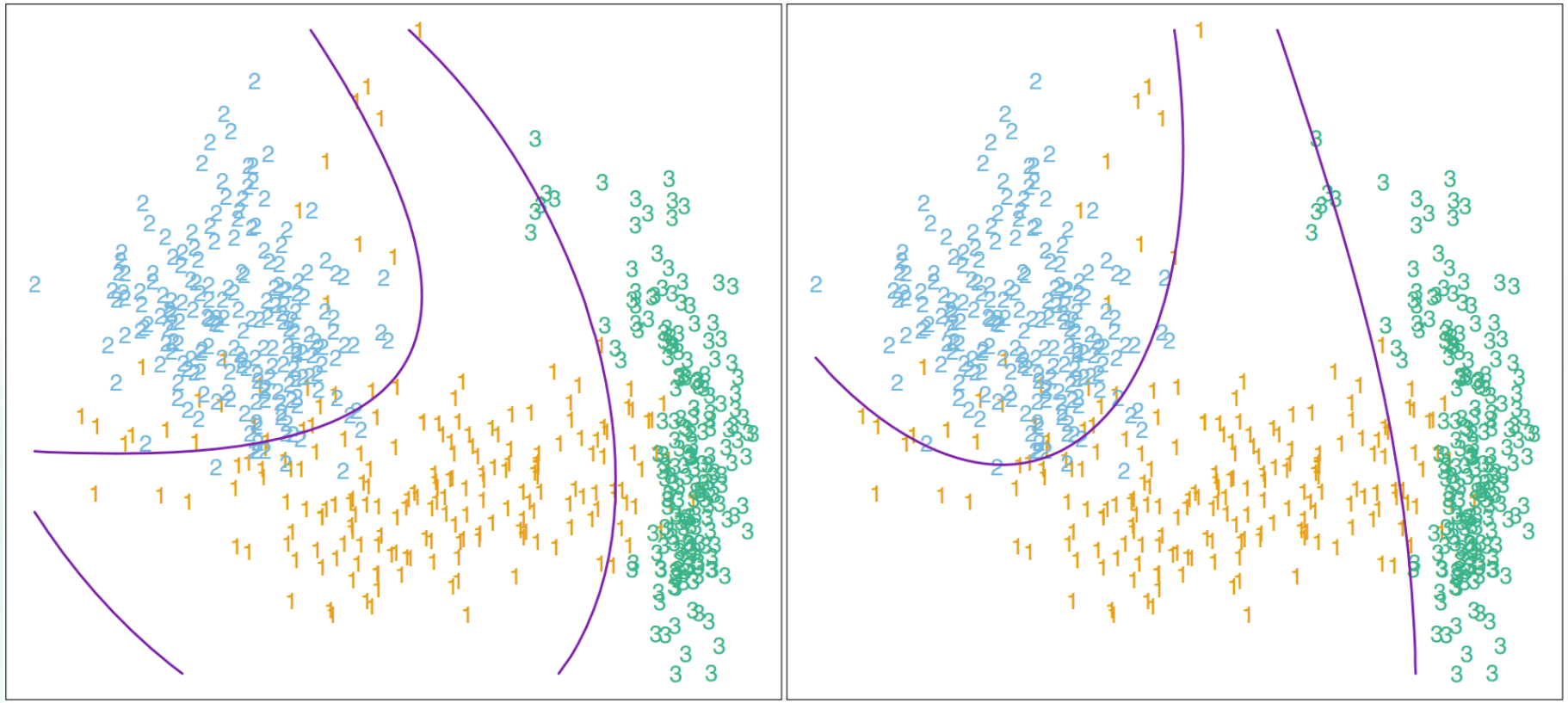
- For QDA, we can use the *quadratic discriminant functions*:

$$\delta_k(x) = \log \pi_k - \frac{1}{2}\log|\Sigma_k| - \frac{1}{2}(x-\mu_k)^T \Sigma_k^{-1}(x-\mu_k).$$

and $\hat{\Sigma}_k, k = 1, \ldots, K$ are estimated from each class.

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Quadratic Discriminant Analysis



- Left: LDA decision boundaries in the five-dimensional space $X_1, X_2, X_1^2, X_2^2, X_1X_2$.
- Right: QDA decision boundaries.

Naive Bayes Classifier

- Again, from the Bayes optimal classifier, it is sufficient to compare

$$\pi_k f_k(x) \text{ or } \pi_k P(x = x|Y = k).$$

However, the joint distribution is usually hard to estimate. Thus one can make a further simplification of conditional independence:

$$f_k(x) = f_k(x_1) f_k(x_2) \cdots f_k(x_p),$$

or

$$P(x|k) = P(x_1|k) P(x_2|k) \cdots P(x_p|k).$$

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Naive Bayes Classifier

- The classification rule becomes

$$\hat{y}(x) = \operatorname{argmax}_k \pi_k \prod_{i=1}^{p} f_k(x_i),$$

or

$$\hat{y}(x) = \operatorname{argmax}_k \pi_k \prod_{i=1}^{p} P(x_i|k).$$

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Example: Text Classification

- Is this spam?

Subject: **Important notice!**
From: Stanford University <newsforum@stanford.edu>
Date: October 28, 2011 12:34:16 PM PDT
To: undisclosed-recipients:;

Greats News!

You can now access the latest news by using the link below to login to Stanford University News Forum.

http://www.123contactform.com/contact-form-StanfordNew1-236335.html

Click on the above link to login for more information about this new exciting forum. You can also copy the above link to your browser bar and login for more information
about the new services.

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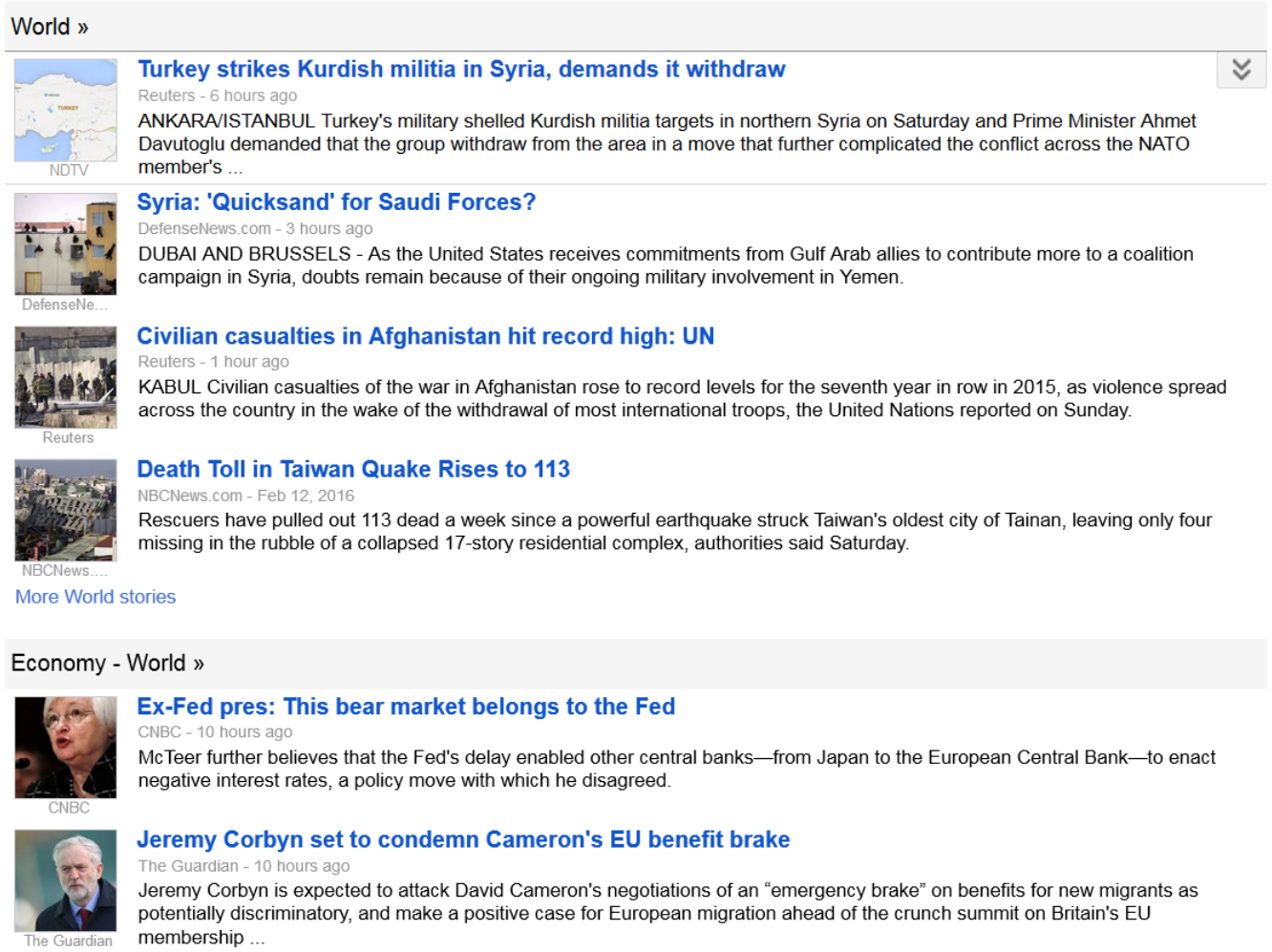
Example: Text Classification

- Positive or negative movie review?
  - “unbelievably disappointing”
  - “Full of zany characters and richly applied satire, and some great plot twists”
  - “this is the greatest screwball comedy ever filmed”
  - “It was pathetic. The worst part about it was the boxing scenes.”

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Example: Text Classification

- What is the category of the news?



Example: Text Classification

- What is the subject of this paper?

MEDLINE Article



Syntactic frame and verb bias in aphasia: Plausibility judgments of undergoer-subject sentences

?

MeSH Subject Category Hierarchy

- Antogonists and Inhibitors
- Blood Supply
- Chemistry
- Drug Therapy
- Embryology
- Epidemiology
- …

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Naive Bayes Text Classifier

- Document $d = \{X_1, X_2, \ldots, X_N\}, X_i \in$ dictionary.

- Conditional independence assumption:

$$\hat{y}(d) = \operatorname{argmax}_k \pi_k \prod_{i=1}^{N} P(X_i|k)\,.$$

Each word follows a categorical distribution.

- Bag-of-words assumption: the word order doesn't matter

$$P(X_i = w^*|k) = P(w^*|k).$$

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Naive Bayes Text Classifier

- To estimate word frequency:

$$\hat{P}(w^*|k) = \frac{count(w^*, k)}{\sum_{w \in D} count(w, k)}.$$

- Problem: what if word w^* did not appear in a particular class in the training set?
- Laplace (add 1) smoothing:

$$\hat{P}(w^*|k) = \frac{count(w^*, k) + 1}{\sum_{w \in D} count(w, k) + |D|}.$$

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Naive Bayes Text Classifier

- In practice, use

$$\hat{y}(d) = \operatorname{argmax}_k \left\{ \log \hat{\pi}_k + \sum_{i=1}^{N} \log \hat{P}(x_i|k) \right\}$$

instead of

$$\hat{y}(d) = \operatorname{argmax}_k \hat{\pi}_k \prod_{i=1}^{N} \hat{P}(x_i|k)$$

to prevent floating-point underflow.

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Naive Bayes Text Classifier

- This text classification model is called *multinomial naive Bayes classifier*.

- Same result if we let $\boldsymbol{x}$ be the |D|-dimensional word count vector and model $P(\boldsymbol{x}|k)$ as a multinomial distribution.

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Naive Bayes Text Classifier – binary version

- Alternatively, for short texts, it is common to use a binary vector $\boldsymbol{x} = (x_1, x_2, \ldots, x_{|D|})$ to indicate if a dictionary word x_j appears in the article.

- In this case we can again assume

$$P(\boldsymbol{x}|k) = \prod_{i=1}^{|D|} P(x_i|k),$$

and model $P(x_i|k)$ with Bernoulli distribution.

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Naive Bayes Text Classifier – TF-IDF version

- Term frequency (TF) of word w: synonymous to word count of w in the document.

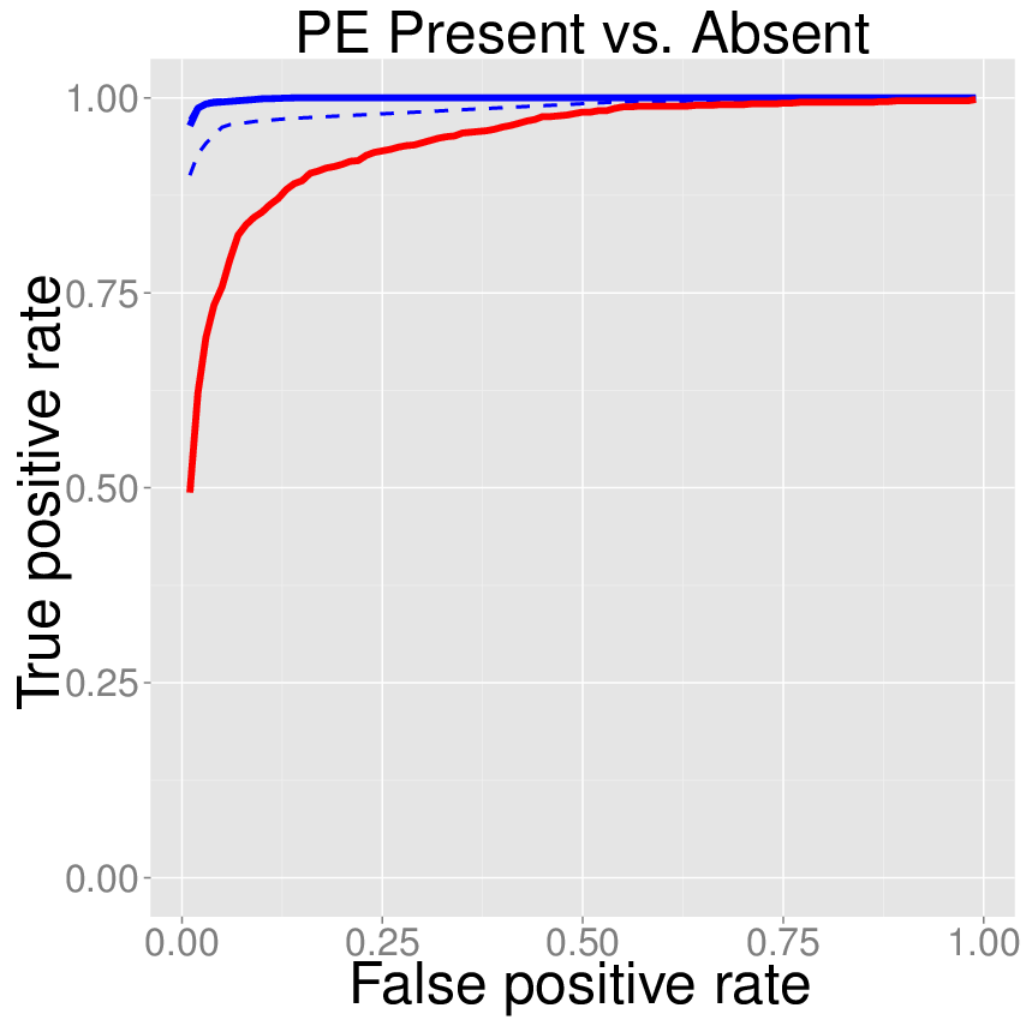
- Inverse document frequency (IDF) of word w: a weight of importance for word w, calculated as:

$$IDF(w) = \log \frac{\text{total \# of documents}}{\text{\# of documents containing } w}$$

- Instead of counting $TF(w)$ times, count $TF(w) \times IDF(w)$ times.

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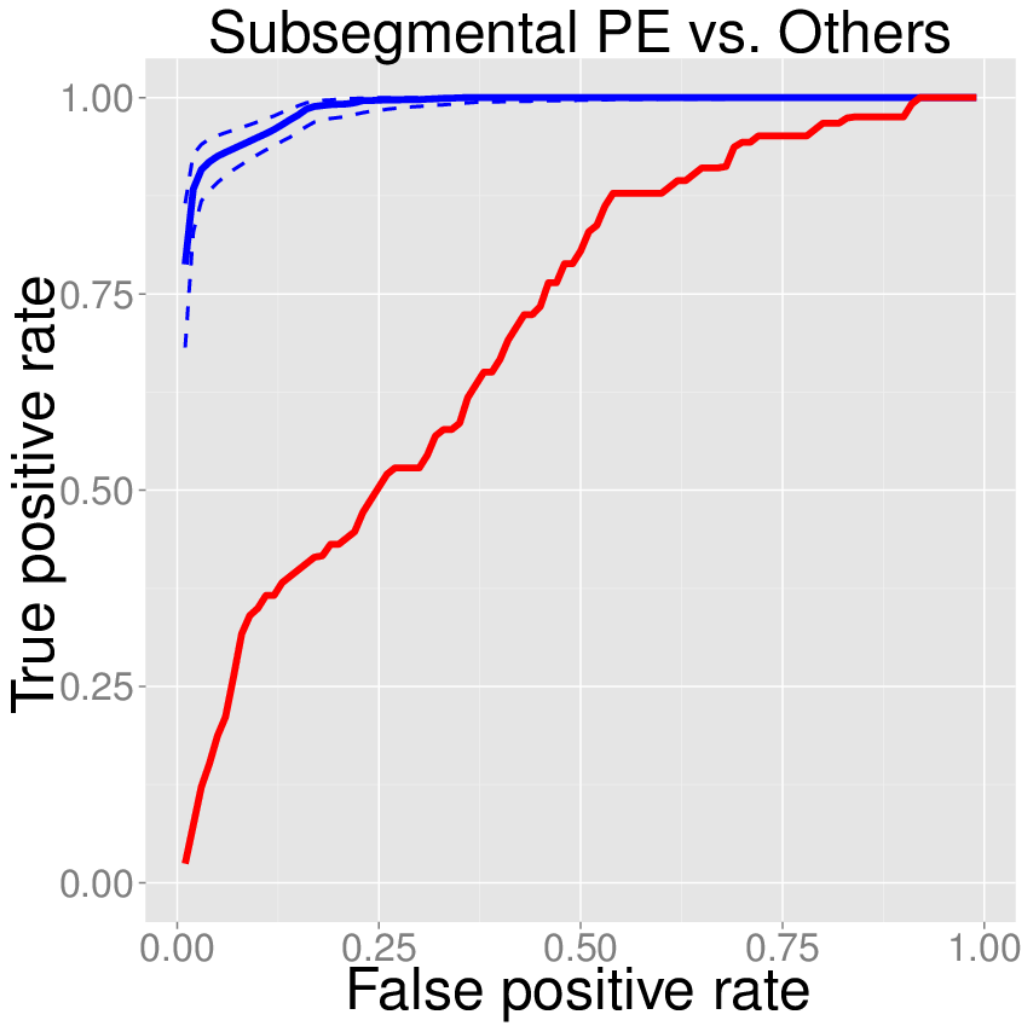
Naive Bayes: simple yet highly performant



Yu et al. (2014) AUC
0.998
NB+BoW AUC
0.942

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Naive Bayes: simple yet highly performant



Yu et al. (2014) AUC
0.986
NB+BoW AUC
0.712

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Logistic Regression

Logistic Regression

- Contrary to Naive Bayes that models $P(X|Y)$, logistic regression models $P(Y|X)$ directly.

- For binary classification,

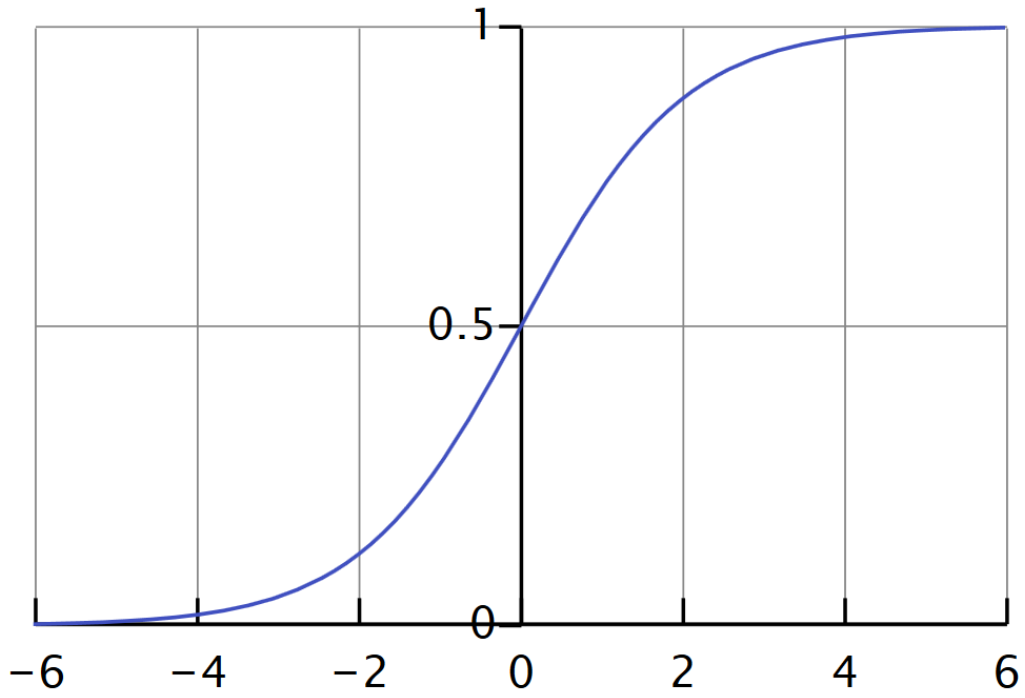
$$P(Y = 1|X = x) = \frac{\exp(\beta_0 + x^T\beta)}{1 + \exp(\beta_0 + x^T\beta)}.$$

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Logistic Regression

- The *logistic* function

$$f(x) = \frac{e^x}{1 + e^x}.$$



Logistic Regression

- For multiple classification,

$$P(Y=k|X=x)=\frac{\exp\left(\beta_{k,0}+x^T\beta_k\right)}{1+\sum_{l=1}^{K-1}\exp\left(\beta_{l,0}+x^T\beta_l\right)},k=1,\dots,K-1,$$

$$P(Y=K|X=x)=\frac{1}{1+\sum_{l=1}^{K-1}\exp\left(\beta_{l,0}+x^T\beta_l\right)}.$$

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Logistic Regression

- Equivalently, logistic regression states that the log-odds (called the *logit* function, which is the inverse of the logistic function) is linear in x.

$$logit(p) = \log\frac{p}{1-p} = \beta_0 + x^T\beta$$

- Odds ratio and e^{β_k} in binary classifications.

- For multiclass classifications, the log-odds of two posterior probabilities are linear functions in x. Particularly,

$$\log\frac{P(Y=k|X=x)}{P(Y=K|X=x)} = \beta_{k,0} + x^T\beta_k, k = 1, \dots, K-1.$$

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Related terms

- Multi-class logistic regression is also known as
  - Maximum entropy or MaxEnt
  - Multinomial logistic regression
  - Softmax regression

- Softmax function:

$$\sigma(\boldsymbol{z})_i = \frac{e^{z_i}}{\sum_{k=1}^{K} e^{z_k}}$$

Fitting Logistic Regression

- Logistic regression models are fit by maximum likelihood. The log-likelihood for the observations is

$$l(\beta) = \sum_{i=1}^{N} \log p_{y_i}(x_i; \beta),$$

where $p_{y_i}(x_i; \beta) = P(Y = y_i | X = x_i; \beta)$.

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Fitting Logistic Regression

- For binary classifications, the calculation simplifies considerably.
- Code y with 0-1, and assume that x includes the constant term 1.

$$l(\beta) = \sum_{i=1}^{N}\{y_i \log p(x_i;\beta) + (1-y_i)\log\big(1-p(x_i;\beta)\big)\}$$

$$= \sum_{i=1}^{N}\{y_i x_i^T\beta - \log\big(1+\exp\big(x_i^T\beta\big)\big)\}.$$

Fitting Logistic Regression

- The log-likelihood is maximized by setting the derivative to zero.

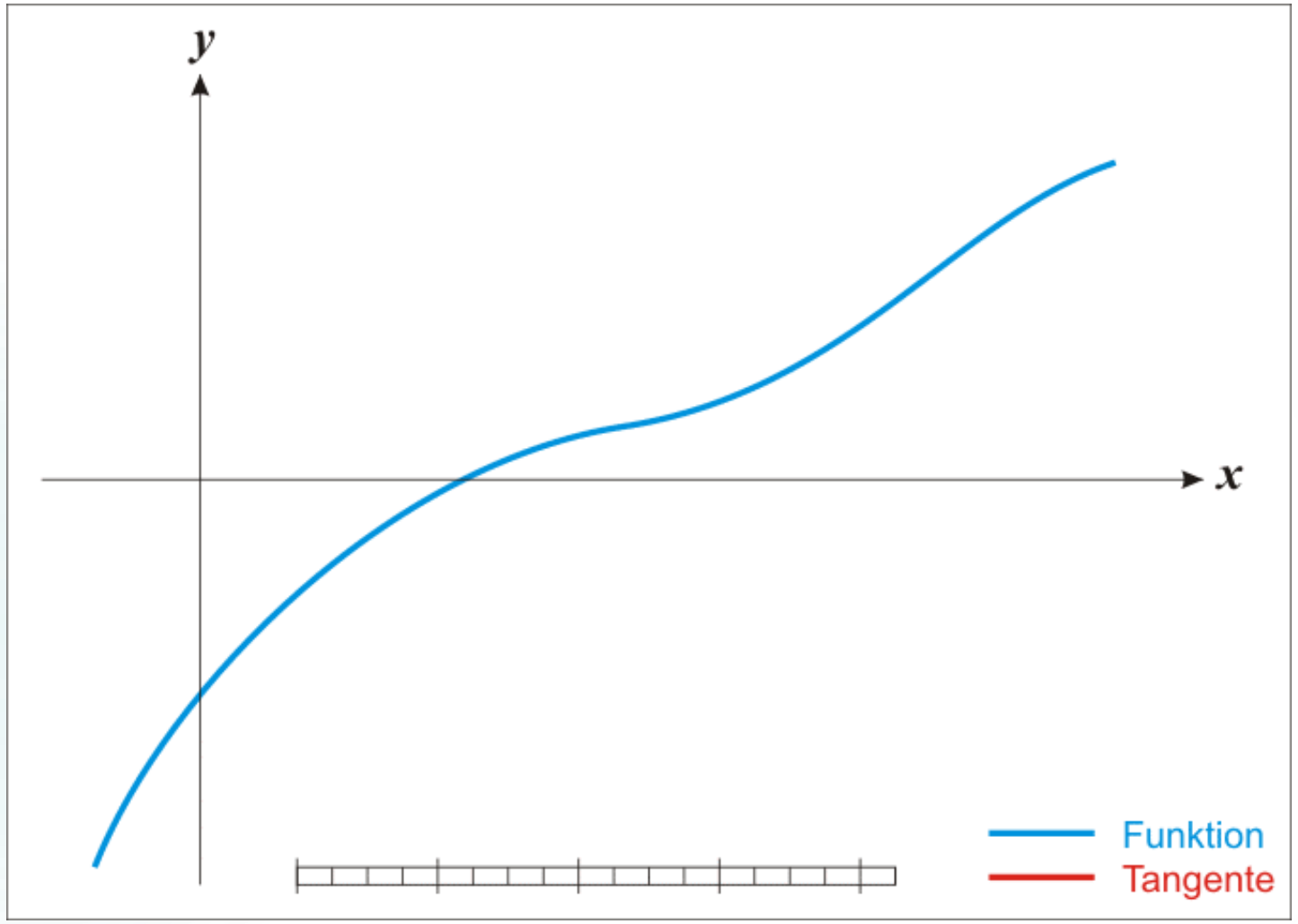
$$\frac{\partial l(\beta)}{\partial \beta} = \sum_{i=1}^{N} x_i \left(y_i - p(x_i;\beta)\right) = X^T(y-p).$$

- Solve it with the Newton-Raphson algorithm

$$x_{\text{new}} = x_{\text{old}} - \frac{f(x_{\text{old}})}{f'(x_{\text{old}})}.$$

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Newton-Raphson Algorithm



Fitting Logistic Regression

$$\beta_{\text{new}} = \beta_{\text{old}} - \left(\frac{\partial^2 l(\beta)}{\partial\beta\partial\beta^T}\right)^{-1}\frac{\partial l(\beta)}{\partial\beta},$$

where

$$\frac{\partial l(\beta)}{\partial\beta} = X^T(y-p),$$

$$\left(\frac{\partial^2 l(\beta)}{\partial\beta\partial\beta^T}\right) = -\sum_{i=1}^{N} x_i x_i^T p(x_i;\beta)\big(1-p(x_i;\beta)\big) = -X^T W X,$$

where $W = diag\{p(1-p)\}$.

Fitting Logistic Regression

- Thus,

$$\beta_{\text{new}} = \beta_{\text{old}} + \left(X^T W X\right)^{-1} X^T (y - p).$$

```
b.new = b.old + solve(t(X) %*% diag(p*(1-p)) %*% X, t(X) %*% (y - p))
b.new = b.old + solve(t(X * (p*(1-p))) %*% X, t(X) %*% (y - p))
```

Multinomial Logistic Regression

$$L(\beta) = \prod_{i=1}^{n} \left(p_1(x_i)\right)^{I(y_i=1)} \cdots \left(p_{\mathrm{K}}(x_i)\right)^{I(y_i=K)}$$

$$l(\beta) = \sum_{i=1}^{N} \sum_{k=1}^{K} I(y_i = k) \log p_k(x_i)$$

$$\frac{\partial l}{\partial \beta_k} = \sum_{i=1}^{N} \left(I(y_i = k) - p_k(x_i)\right) \cdot x_i$$

$$\frac{\partial^2 l}{\partial \beta_k \partial \beta_k'} = -X' \cdot diag\left(p_k(x_i)\left(1 - p_k(x_i)\right)\right) \cdot X$$

$$\frac{\partial^2 l}{\partial \beta_k \partial \beta_j'} = -X' \cdot diag\left(p_k(x_i) p_j(x_i)\right) \cdot X$$

R Packages


```
library(glmnet)
glmnet(x,y,family="binomial") # two-class logistic regression
glmnet(x,y,family="multinomial") # multi-class logistic regression
```

Regularized Logistic Regression

- L_2 penalty
- L_1 penalty
- Elastic-Net penalty
- Etc.

```
glmnet(x, y, family="binomial", alpha, lambda)
cv.glmnet
```

Logistic Regression with TF-IDF?

- What does it mean?
- When does it make sense?

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Question: sentiment analysis

- Positive or negative movie review?
  - “unbelievably disappointing”
  - “Full of zany characters and richly applied satire, and some great plot twists”
  - “this is the greatest screwball comedy ever filmed”
  - “It was pathetic. The worst part about it was the boxing scenes.”

- For sentiment analysis, which one do you think is better? Naïve Bayes or logistic regression?

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Summary Points

- Logistic regression generally performs a little better than Naïve Bayes.
- Logistic regression is commonly used in biomedical studies for interpretability.
- Logistic regression should be prioritized when the sample size is small.

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Model Selection

Akaike Information Criterion

- $$\mathrm{AIC} = -2 \cdot \mathrm{loglik} + 2d,$$

  where d is degrees of freedom (df).

- Not so helpful definitions:
  - The degrees of freedom is the number of values in the final calculation of a statistic that are free to vary.
  - The degrees of freedom is the number of independent ways by which a dynamic system can move, without violating any constraint imposed on it. In other words, the number of degrees of freedom can be defined as the minimum number of independent coordinates that can specify the position of the system completely.

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Common Degrees of Freedom

- OLS: $p + 1$
- Ridge: $\mathrm{tr}\left(X\left(X^T X + \lambda I\right)^{-1} X^T\right)$
- Lasso: the number of active features (i.e., nonzero coefficients) in the fitted model
- ENet: $\mathrm{tr}\left(X_A\left(X_A^T X_A + \lambda_2 I\right)^{-1} X_A^T\right)$, where A is the active set.

Log-likelihood

- Logistic regression: binomial/multinomial likelihood.
- Linear regression: Gaussian assumption on the noise term.

$$\text{AIC} = \frac{\text{RSS}}{\hat{\sigma}^2} + 2d,$$

where $\hat{\sigma}^2$ is estimated from a low-bias model, such as the OLS using the full model.

Motivation of the AIC (optional)

- Kullback-Leibler divergence:

$$D_{KL}(f||g(\cdot|\theta)) = \int_{\Omega} f(x) \log\left(\frac{f(x)}{g(x|\theta)}\right) \mathrm{d}x$$
$$= \int_{\Omega} f(x) \log f(x) \, \mathrm{d}x - \int_{\Omega} f(x) \log g(x|\theta) \, \mathrm{d}x$$
$$= -H(f) + H(f, g)$$

- f: truth in terms of a probability distribution. x: the outcome.
- g: approximating model in terms of a probability distribution.
- θ: parameter vector in the approximating model g.
- $H(f)$: entropy of the distribution f.
- $H(f, g)$: cross-entropy of the distributions f and g.
- $D_{KL}(f||g) \geq 0$, with $D_{KL}(f||g) = 0$ if and only if $f = g$ almost everywhere.

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Motivation of the AIC (optional)

- Given observation y and let $\hat{\theta}(y)$ be the MLE estimate based on model g and observation y, we want to minimized the expected KL divergence

$$\mathbb{E}_y\left(D_{KL}\left(f||g\left(\cdot|\hat{\theta}(y)\right)\right)\right)$$

$$= \int_\Omega f(x)\log f(x)\,\mathrm{d}x - \int_\Omega f(y)\int_\Omega f(x)\log g(x|\hat{\theta}(y))\,\mathrm{d}x\,\mathrm{d}y$$

$$= \int_\Omega f(x)\log f(x)\,\mathrm{d}x - \mathbb{E}_y\mathbb{E}_x \log g\left(x|\hat{\theta}(y)\right)$$

- Asymptotically, $\mathbb{E}_y\mathbb{E}_x \log g\left(x|\hat{\theta}(y)\right) = \text{loglik} - d$.

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Bayesian Information Criterion

- $$\mathrm{BIC} = -2 \cdot \mathrm{loglik} + (\log N) \cdot d,$$

where d is degrees of freedom (df).

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Motivation of the BIC (optional)

- Suppose that we have a set of candidate models $\mathcal{M}_m, m = 1, \ldots, M$ and corresponding model parameters θ_m. Note: different penalty parameters would indicate different models.
- Let $\boldsymbol{Z}$ be the observed data. From a Bayesian perspective, we choose the model that has the largest the posterior probability $P(\mathcal{M}_m|\boldsymbol{Z})$
- To compare models $\mathcal{M}_m$ and $\mathcal{M}_l$, it is sufficient to compare their log-odds:

$$\log\frac{P(\mathcal{M}_m|\boldsymbol{Z})}{P(\mathcal{M}_l|\boldsymbol{Z})} = \log\frac{P(\mathcal{M}_m)}{P(\mathcal{M}_l)} + \log\frac{P(\boldsymbol{Z}|\mathcal{M}_m)}{P(\boldsymbol{Z}|\mathcal{M}_l)}.$$

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Motivation of the BIC (optional)

- Since we typically have no knowledge about the prior distribution of the model, we assume a uniform distribution, i.e., $P(\mathcal{M}_m) = P(\mathcal{M}_l)$.
- Using Taylor expansion and Laplace approximation,

$$\log P(\boldsymbol{Z}|\mathcal{M}_m) \approx \log P\left(\boldsymbol{Z}|\hat{\theta}_m, \mathcal{M}_m\right) - \frac{1}{2} d_m \cdot \log N,$$

where $\hat{\theta}_m$ is a maximum likelihood estimate and d_m is the degrees of freedom in model $\mathcal{M}_m$.

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Comparisons between AIC and BIC

- The AIC penalizes the number of parameters less strongly than does the BIC.
- Burnham & Anderson (2002): AIC can be derived in the same Bayesian framework as BIC, just by using a different prior.
- AIC is derived from principles of information; BIC is not.
- Yang (2005): AIC is asymptotically optimal in selecting the model with the least mean squared error. The rate at which AIC converges to the optimum is the best possible.